The Risk-Bearing Subject of AI-Related Existential Risk to Humanity

From Species Survival to the Realization of Subjecthood

under Civilizational Conditions

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Abstract

AI safety, risk, and governance are becoming global agenda items, but the intensification of governance activity does not automatically clarify the object of risk. When different fields say that they seek to protect "humanity," they implicitly point to preferences, lives, social order, or species continuity—concepts that are not mutually compatible. As a result, risk identification continually slides across dimensions. This paper is an exercise in conceptual engineering concerning the **risk-bearing subject** of AI-related existential risk to humanity. It also has significance for **social epistemology**, the philosophy of technology, and AI risk governance.

This paper preserves the baseline of existing frameworks that understand "humanity" as *Homo sapiens*, but proposes a paradigm extension: the risk-bearing subject is always humanity as an **embodied species**; what this paper extends is not the risk-bearing subject, but the ways in which one and the same subject may suffer existential harm. The paper distinguishes two basic layers of possible harm—the **biological-survival layer** and the **civilizational-conditions layer**—and, within the latter, identifies the **philosophical kernel**, namely the **category-calibration mechanism** of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. AI may, through **deprivation by substitution** or **mediated reconfiguration**, weaken humans' **substantive participation** in the functional systems of civilization, or cause foundational categories to lose stable calibration. Risk to the realization of subjecthood at the civilizational-conditions layer enters the scope of existential risk only when it acts upon foundational civilizational conditions, has diffusion capacity across domains and civilizational interfaces, may cause irreversible or difficult-to-reverse structural damage, and causes these harms to converge into overall, intergenerational damage to humanity's capacity **as a future-oriented subject**.

What this paper provides is not a complete taxonomy of risks or a governance scheme, but the **ontological coordinate system** needed for AI risk identification. The central point is not to add civilization, institutions, or the philosophical kernel as new risk-bearing subjects, but to show how the same human subject may suffer existential harm through damage to civilizational conditions.

Keywords: artificial intelligence; existential risk to humanity; risk-bearing subject; conceptual engineering; realization of subjecthood; category calibration

1 | Introduction: AI Reopens the Problem of Existential Risk to Humanity

1.1 The Problem: Intensified Governance and the Absence of Ontological Coordinates

Artificial intelligence is reopening the problem of existential risk to humanity. The risks discussed in this paper are situated against the background of the **AGI era** [Note 1]. By the AGI era, this paper does not mean that any widely accepted AGI threshold has already been crossed. It refers instead to an historical transition in which AI systems are becoming more general, more autonomous, more capable of action, and more deeply embedded in institutions. Many of the risk mechanisms discussed in this paper may already be visible at the current stage in the form of advanced AI, generative AI, and agentic AI.

Model evaluations, safety commitments, risk classifications, compliance frameworks, and international coordination mechanisms continue to proliferate. AI safety has rapidly moved from a topic discussed by a relatively small technical community to one shared by governments, international organizations, frontier laboratories, and corporations. These efforts show that humanity has become aware that AI may reshape the basic structures of knowledge, institutions, the economy, and security [Note 2]. Yet the intensification of governance activity does not automatically clarify the object of risk.

Different institutions understand the “humanity” to be protected from their own positions: corporations focus on user safety and product liability; governments focus on national security and public order; research institutions focus on model capabilities and system reliability; ethical discussions focus on bias, fairness, rights, and transparency. These responses operate at the application layer, institutional layer, or governance layer. They presuppose some human interest to be protected, but they do not necessarily share an ontological coordinate for **humanity as the risk-bearing subject**.

When AI safety, risk analysis, and governance all say that they aim to protect “humanity,” they do not always first specify what “humanity” means: a biological species, an aggregate of preferences, a population of users, a civic order, or an embodied species realizing subjecthood under civilizational conditions. If this risk-bearing subject is not clarified, governance may expand through continual repair at the model, product, and policy levels, while still lacking the conceptual basis for judging which kinds of harm touch the foundations of human existence.

1.2 The Sliding Concept of “Humanity” in Risk Discussions

Current discussions of AI risk implicitly rely on different concepts of “humanity.” Alignment research typically assumes that what is to be protected is human preference; biosecurity research assumes that it is human life; governance research assumes that it is social order; philosophical discussion assumes that it is human value and dignity; and the classical framework of existential risk, while not limited to species extinction and while also including permanent incapacitation and future deprivation, still typically takes humanity as a biological species and its long-term future as its basic point of reference [Note 3]. In our view, these different concepts of “humanity” are not compatible. When they are mixed within the same policy document or technical roadmap, risk identification continues to slide across dimensions: at one moment the discussion concerns an epistemic issue, such as whether a model outputs what is true; at the next it turns to biosecurity, such as whether a model may be misused; and at the next it becomes a matter of economic adaptation, such as whether jobs will be replaced.

If the risk-bearing subject remains unclear, different lines of research will never have a shareable ontological coordinate.

1.3 Core Claim and Methodological Position

The core claim of this paper is the following: **the risk-bearing subject of AI-related existential risk to humanity is always humanity as an embodied species; this paper does not extend the risk-bearing subject to civilization, institutions, the philosophical kernel, or AI systems.** What it extends is the range of ways in which one and the same subject may suffer existential harm: from destruction, permanent incapacitation, or future deprivation at the biological-survival layer alone, to the irreversible or difficult-to-reverse weakening of the foundational conditions under which humanity realizes its subjecthood under civilizational conditions.

Accordingly, this paper does not deny that existing existential-risk frameworks already include the dimension of "future deprivation." Its contribution is to point out that, in the age of AI, future deprivation cannot be understood only as a consequence attached to biological extinction. It may also appear in the form of humanity continuing to exist while gradually losing the civilizational conditions under which it can understand, judge, take responsibility, and shape the common world as a future-oriented subject.

To this end, the paper undertakes an exercise in **conceptual engineering** [Note 4]. It distinguishes two basic layers of possible harm to humanity as the risk-bearing subject: the biological-survival layer and the civilizational-conditions layer. The former concerns the survival of humanity as an embodied species; the latter concerns humanity's capacity, within an intergenerational common world, to know what is true, form judgment, bear responsibility, participate in institutions, create meaning, and act toward the future. Within the civilizational-conditions layer, the paper further identifies the **philosophical kernel**, namely the **category-calibration mechanism** of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. The philosophical kernel is not an independent entity, but the mechanism within civilizational conditions that maintains the operability of foundational categories.

What this paper provides is not a complete taxonomy of risks, nor a technical-control scheme or a policy checklist. It provides the **ontological coordinate system** needed at the level of risk identification in AI safety, risk analysis, and governance: before discussing concrete risk types, governance pathways, or technical solutions, we must first clarify "humanity" as the risk-bearing subject and the ways in which it may be harmed.

1.4 Structure of the Argument

Section 2 offers a two-dimensional working definition of humanity as an embodied species. Section 3 defines human civilization as the intergenerational condition for the realization of subjecthood and distinguishes **formal participation** from **substantive participation**. Section 4 argues that the philosophical kernel is a category-calibration mechanism and establishes the mapping relation among foundational categories, foundational civilizational conditions, and capacities of subjecthood. Section 5 uses a limited functional analogy to explain the relations among the four levels and then restates their ontological relation in non-analogical terms. Section 6 explains why AI makes these conditions an issue of existential risk, and proposes two mechanisms of action, an expanded definition, and four thresholds. Section 7 responds to objections through test cases and engagement with adjacent literature. Section 8 concludes the paper.

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2 | Humanity as an Embodied Species: A Two-Dimensional Working Definition

If "humanity" is understood only as *Homo sapiens*, a biological species, then existential risk will primarily refer to species extinction, permanent incapacitation, and future deprivation. This understanding captures the most basic form of risk, but it is not sufficient to exhaust the object of human risk in the age of AI.

2.1 A Two-Dimensional Working Definition of Humanity

In this paper, "humanity" refers first to *Homo sapiens* as an embodied biological population. Humanity has bodily structures, nervous systems, perceptual capacities, reproductive mechanisms, physiological vulnerability, mortality, and dependence on ecological and material environments. This **biological-survival layer** constitutes the physical precondition of human existence.

Yet humanity does not exist only at the biological-survival layer. Humanity as an embodied species also realizes **subjecthood** under **civilizational conditions**: through language, knowledge, education, institutions, structures of responsibility, public reason, historical memory, and narratives of meaning, humans acquire and sustain the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future.

Thus, the working definition of "humanity" in this paper has two layers. First, humanity is an embodied biological species. Second, humanity is an embodied species that realizes subjecthood under civilizational conditions. The former identifies the biological bearer of human existence; the latter explains why humanity is not merely a biological population, but a historical form of existence capable of knowing, judging, taking responsibility, transmitting, creating meaning, and acting toward the future.

The risk-bearing subject is always humanity as an embodied species. Civilization, institutions, cultures, technological systems, and the philosophical kernel are not risk-bearing subjects alongside humanity. What this paper extends is not the risk-bearing subject itself, but the ways in which one and the same subject may suffer existential harm.

2.2 Species Extinction as Baseline Risk

Human species extinction is the most direct and strictest form of existential risk to humanity. If *Homo sapiens* ceases to exist in the biological sense, or if humanity permanently loses the capacity to reproduce, act, and shape the future, then every possibility concerning human civilization and the human future is closed. Without embodied human life, there is no continuing unfolding of human civilization and no bearer of subjecthood-realization.

Species survival is the baseline that any expanded definition must preserve. It is a necessary condition, but not a sufficient condition. Humanity may continue to exist biologically while gradually losing, in a civilizational sense, the foundational capacities that allow it to continue existing as knower, judge, bearer of responsibility, institutional participant, and creator of meaning. In other words, species survival answers the question "Are there still humans?" Civilizational conditions answer the question "Can humanity still exist as the subject of its own future?"

2.3 Embodiment: Humanity Is Not a Pure Information Structure

To emphasize humanity as an embodied species is to refuse to reduce humanity to a pure information structure, an abstract rational subject, or a computational pattern that can be arbitrarily migrated. Human cognition, affect, judgment, learning, relationships, responsibility, and meaning are all closely tied to bodily conditions. Perceiving the world depends on the body; language learning depends on embodied experience and social interaction; bearing responsibility depends on the capacity to act and to undergo consequences; the formation of meaning is also tied to birth, death, time, suffering, dependence, relationships, and finitude.

In the context of AI, this point must be preserved with particular care. As AI systems increasingly perform cognitive tasks, generate texts, organize knowledge, propose plans, and participate in decision-making, humans are easily understood as cognitive entities that can be replaced by more efficient information-processing systems. Yet human existence is not merely information processing. A being that can die, be injured, make promises, feel regret, grow through time, and bear consequences does not share the same structure of existence as an information system without a body, mortality, or vulnerability.

2.4 Species Continuity Is Not the Same as Continuity of Civilizational Conditions

The continuity of the human species is maintained primarily through biological reproduction, intergenerational succession, and population stability. As long as humans can still reproduce, survive, and maintain their basic biological structure, continuity at the species level remains. The continuity of civilizational conditions is different. It depends on the transmission of language, memory, education, institutions, knowledge, norms, technology, structures of responsibility, and narratives of meaning. It requires not only that the next generation be born, but also that the next generation enter a common world that can be understood, inherited, criticized, and renewed.

There is therefore no automatic equivalence between species continuity and the continuity of civilizational conditions. A society may continue biologically while undergoing a severe rupture in civilizational conditions. People may still be born, work, consume, and entertain themselves, but the common world into which they enter may no longer be able to stably maintain knowledge, responsibility, judgment, and the transmission of meaning. If such degradation is irreversible or difficult to reverse, it is not merely ordinary social change. It may become structural damage to the conditions for the realization of human subjecthood.

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3 | Human Civilization: The Intergenerational Conditions for the Realization of Subjecthood

3.1 Civilization and Human Civilization: Scope and Limitation

In this paper, "civilization" always means "human civilization." Human civilization is the **intergenerational common world** formed by humanity, as an embodied species, through language, knowledge, institutions, law, education, structures of responsibility, technological practices, and narratives of meaning. It consists of transmissible, revisable, and inheritable symbolic-institutional-technical structures that enable individuals to enter a historical and interpretive space larger than their biological lives.

Human civilization is not an independent risk-bearing subject outside humanity. It is a **constitutive condition** for the realization of human subjecthood. It is not the simple sum of population, states, culture, or technological systems.

3.2 Four Negations

Human civilization is not the sum of population. The simultaneous existence of many human individuals does not automatically constitute civilization. Population provides the biological support of civilization, but population alone does not automatically generate transmissible knowledge, institutions, responsibility, and structures of meaning.

Human civilization is not the state. The state is a form of political and legal organization, typically involving territory, sovereignty, and coercive power. Human civilization is a deeper, broader, and longer-term intergenerational common world. A civilizational tradition may span multiple states, and a state may carry multiple civilizational traditions.

Human civilization is not synonymous with culture. Culture is an important component of civilization, appearing in language, customs, art, ritual, and ways of life. But civilization also includes science, law, institutions, education, the economy, public reason, infrastructure, and structures of responsibility.

Human civilization is not a technological system. Technology is an important component of civilization, but civilization is not merely a collection of tools, nor is it a system of efficiency. Civilization also includes value, norms, responsibility, public reason, meaning, and humanity's understanding of the future. Growth in capability is not automatically equivalent to the health of civilizational conditions.

3.3 Multiple Civilizational Traditions and the Modern Global Civilizational Ecology

Human civilization should be understood as an overall civilizational ecology composed of multiple civilizational traditions, shared institutional interfaces, knowledge exchange, technological infrastructures, experiences of conflict, and structures of intergenerational transmission. Different civilizational traditions may give different answers to questions of truth, responsibility, value, subjecthood, and meaning. Yet together they constitute the intergenerational common world formed and inherited by humanity as an embodied species throughout history.

The modern global civilizational ecology has produced an unprecedented degree of mutual embeddedness among different civilizational traditions. Modern scientific systems, university institutions, international law, global trade, financial networks, Internet protocols, digital platforms, and AI infrastructures no longer belong entirely within any single civilizational tradition, state, or culture. They constitute shared interfaces that operate across civilizations. Thus, the civilizational-conditions risks discussed in this paper do not take local disorder within a single state, culture, or institutional system as sufficient. They refer instead to structural harms that may act upon these shared cross-civilizational interfaces.

3.4 Functional Systems of Civilization and Structures of Human Participation

Human civilization operates concretely through a set of **functional systems of civilization**. Economic, legal, educational, scientific, state-governance, cultural, media, medical, administrative, religious, and artistic systems constitute the main pathways through which humans actually enter civilizational conditions. Ordinary people usually do not encounter "civilization" as an abstract object. They enter the common world through schools, courts, laboratories, markets, media, government agencies, platform systems, families, public services, and cultural practices.

These functional systems are not external appendages to civilization. They are important support structures for the realization of human subjecthood. Through educational systems, humans learn how to understand, question, judge, and inherit knowledge; through scientific systems, they participate in knowledge production and the testing of evidence; through legal systems, they understand rights, obligations, and the attribution of responsibility; through governance systems, they participate in public decision-making and institutional direction; through cultural and media systems, they express meaning, transmit memory, and form public imagination.

Human subjecthood does not occur only inside the individual. It is also realized through **structures of participation** within these systems. Structures of participation refer to the ways in which humans participate in the operation, maintenance, and renewal of the common world as learners, workers, judges, creators, citizens, bearers of responsibility, and future-oriented subjects.

3.5 Formal Participation and Substantive Participation

To analyze structures of participation more precisely, this paper distinguishes **formal participation** from **substantive participation** [Note 5].

Formal participation refers to a mode of participation in which humans retain visible positions, procedural actions, and nominal roles within the functional systems of civilization: clicking to confirm, signing documents, approving recommendations, choosing options, voting, consuming, and providing feedback. Formal participation preserves the recognizability of humans within institutions, but it does not guarantee that humans genuinely understand, judge, or shape the process.

Substantive participation refers to a mode of participation in which humans, as knowers, judges, bearers of responsibility, maintainers of institutions, creators of meaning, and future-oriented subjects, exert substantive influence on the operation, direction, and renewal of the common world. It requires that humans be able to understand processes, form judgment, bear responsibility, create meaning, and exert substantive influence on institutional direction and the public future.

If humans retain only formal confirmation within the functional systems of civilization, while no longer genuinely understanding, judging, taking responsibility, creating, or shaping direction, then the realization of human subjecthood under civilizational conditions is weakened.

3.6 The Intergenerational Common World and Humanity as a Future-Oriented Subject

Human civilization is an **intergenerational common world** [Note 6]. Through language, memory, institutions, and education, it enables the past to be transmitted to the future and enables finite individual lives to enter a historical structure larger than themselves. A person is not born from nothing. They enter an already existing language, receive existing institutions and knowledge, and inherit historical memories, scientific achievements, legal structures, ethical norms, and narratives of meaning that they did not personally create.

This intergenerationality makes civilizational conditions vulnerable in a way different from species continuity. Species continuity depends primarily on biological reproduction; the continuity of civilizational conditions depends on transmission, understanding, and renewal. If the next generation is merely born biologically, but does not enter a common world capable of cultivating judgment, responsibility, truth, and meaning, then the conditions for the realization of human subjecthood are weakened.

The true continuity of civilizational conditions is not merely that "some humans remain alive," but that "some humans remain capable of understanding, inheriting, criticizing, and renewing a common world."

At this point, a key term must be established: **humanity as a future-oriented subject**.

This does not introduce a new risk-bearing subject. It refers to the temporal and agential dimension of subjecthood-realization: humanity not only knows, judges, and takes responsibility in the present, but also, across generations, can understand risks, revise institutions, organize action, and shape the future.

3.7 Structures of Knowledge, Responsibility, and Meaning

As an intergenerational common world, human civilization contains at least three functional structures: **structures of knowledge**, **structures of responsibility**, and **structures of meaning**.

Structures of knowledge enable humans to distinguish fact from rumor, evidence from narrative, explanation from fabrication, and knowledge from opinion. Structures of responsibility enable humans to identify actors, trace consequences, allocate obligations, address wrongdoing, and maintain trust. Structures of meaning enable humans to understand why to learn, why to work, why to create, why to bear responsibility, and why to protect the future.

Together, structures of knowledge, responsibility, and meaning show that human civilization is not the simple aggregation of population, states, cultures, or technological systems. It is the common world that enables humans to know, judge, take responsibility, transmit, and create meaning. If these structures are systematically weakened, the conditions for the realization of human subjecthood are damaged.

These structures, in turn, depend on more basic foundational categories and their calibration mechanisms. The next section asks: which foundational categories must the functional systems of civilization invoke? How are these categories clarified, maintained, and adjusted?

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4 | The Philosophical Kernel: The Category-Calibration Mechanism within Civilizational Conditions

4.1 The Definition and Scope of the Philosophical Kernel

The **philosophical kernel** in this paper is not the collection of foundational categories themselves, but the **category-calibration mechanism** within civilizational conditions that maintains the operability of foundational categories. It has the character of a meta-mechanism: here "meta" does not mean that it constitutes a higher entity outside civilizational conditions, nor that it becomes a new risk-bearing subject. It means that what it calibrates is not a local rule internal to one functional system of civilization, but the **foundational category interfaces** jointly invoked by functional systems such as science, law, education, politics, economy, and culture.

Here, "philosophy" does not primarily mean philosophy as an academic discipline, nor only theoretical texts written by professional philosophers. The philosophical kernel refers to the operative mechanism through which foundational categories are made explicit, tested, clarified, criticized, maintained, and adjusted in history; truth, evidence, responsibility, subjecthood, value, and meaning are its objects of calibration and conceptual interfaces.

The philosophical kernel is not an independent entity within civilization, nor an ordinary department parallel to science, law, politics, education, economy, culture, and art. The risk-bearing subject is always humanity as an embodied species; civilizational conditions are the intergenerational common world in which human subjecthood is realized; the philosophical kernel is the category-calibration mechanism within civilizational conditions that maintains the operability of foundational categories.

4.2 Foundational Categories and the Modern Global Civilizational Ecology

Foundational categories are not eternal frameworks that apply a priori to every possible civilization, nor are they theoretical vocabularies monopolized by any one civilizational tradition. Different civilizational traditions may use different languages, religions, rites and laws, institutions, political imaginaries, and knowledge traditions to organize questions of truth, responsibility, subjecthood, value, and meaning. Yet insofar as a civilization can distinguish truth from falsehood, accumulate knowledge, allocate responsibility, organize action, evaluate value, and interpret meaning, it is already invoking some configuration of foundational categories.

The modern global civilizational ecology gives these categories a new commonality. Scientific knowledge, legal responsibility, technological governance, public decision-making, educational transmission, digital platforms, international institutions, and transnational markets are deeply interwoven. A knowledge institution within one state or civilizational tradition often depends on global research networks, technical standards, digital infrastructures, and international institutional interfaces; the deployment of an AI system in one region may also affect other regions through platforms, supply chains, data ecologies, and structures of public discourse.

Concern for the calibrability of truth, evidence, and knowledge structures is not merely a matter of traditional epistemology. It is a central concern of **social epistemology** [Note 7]: knowledge is not the private possession of an individual mind, but a public achievement across subjects, institutions, and generations. It is in this context that truth, evidence, responsibility, subjecthood, value, and meaning become unavoidable foundational categories in AI risk analysis.

This does not mean that different civilizational traditions understand these categories in exactly the same way. Truth may be organized through different knowledge traditions; responsibility may be allocated through different legal and ethical systems; subjecthood may be understood through different conceptions of personhood, family, community, and politics; value and meaning may also be interpreted through different religious, philosophical, cultural, and historical experiences. The argument of this paper is not to erase this diversity, but to point out that, whatever their specific answers, the knowledge, institutional, technological, and governance systems of the modern global civilizational ecology must invoke these interfaces in some form.

Foundational categories are therefore neither purely a priori nor arbitrarily relative. They have **functional necessity**: without truth and evidence, knowledge cannot accumulate stably; without responsibility, action cannot be attributed; without subjecthood, judgment, consent, authorization, and responsibility-bearing lose their foundation; without value, technological optimization loses its goal; without meaning, learning, creation, responsibility, and commitment to the future lose their structure of reasons.

4.3 The Functional Positions of the Six Foundational Categories

This paper focuses on six foundational categories: truth, evidence, responsibility, subjecthood, value, and meaning. They are not the whole set of foundational concepts in human civilization, nor do they claim to exhaust all possible core civilizational categories. They are selected because they constitute the minimal conditions for AI risk identification itself: without truth and evidence, risk cannot be known; without responsibility and subjecthood, action cannot be attributed; without value and meaning, risk governance cannot state what it is protecting or why the future matters.

The category of **truth** enables civilization to distinguish truth from falsehood, fact from narrative, knowledge from opinion, and explanation from fabrication. Without the category of truth, knowledge production becomes content production, and public discussion loses a shared point of reference.

The category of **evidence** answers what can support a judgment, what can refute a judgment, and what can enter the public exchange of reasons. Without the category of evidence, science, law, education, media, and public governance all struggle to distinguish the reliable from the unreliable.

The category of **responsibility** connects action, subject, authorization, intention, fault, consequence, and obligation. Without the category of responsibility, society may still produce outcomes and consequences, but it becomes increasingly difficult to ask who acted, who authorized, who knew, who could have prevented harm, and who ought to bear the consequences.

The category of **subjecthood** answers who can judge, consent, refuse, authorize, act, bear responsibility, and represent themselves. Without the category of subjecthood, law, politics, education, ethics, and institutional participation lose their foundation.

The category of **value** enables civilization to order safety, freedom, efficiency, truth, dignity, justice, innovation, stability, prosperity, and meaning. Without the category of value, technological systems can continue to optimize, but civilization cannot explain what optimization is for.

The category of **meaning** answers why to learn, why to create, why to bear responsibility, why to protect the future, and why humanity still matters. Without the category of meaning, society can continue to produce, consume, entertain, and optimize, but human action, responsibility, and future commitment increasingly lack a shared explanation.

Together, these six foundational categories constitute the underlying interfaces that the functional systems of civilization must invoke. They are not isolated words, but a mutually supporting **category network**. Truth requires evidence for calibration; responsibility requires subjecthood as its bearer; value requires meaning for interpretation; and meaning also requires truth, responsibility, and value if it is not to slide into arbitrary narrative. If multiple foundational categories become miscalibrated at the same time, their mutual support relations may undergo a linked **systemic drift**. The function of the philosophical kernel is precisely to maintain the distinguishability, connectivity, and corrigibility of these categories.

4.4 How the Category-Calibration Mechanism Works

The category-calibration mechanism refers to the process through which foundational categories are made explicit, tested, criticized, maintained, and adjusted in history. It does not imply that a single center can manage all categories, nor that any one discipline can monopolize the authority to interpret them. It means that, under civilizational conditions, foundational categories can be continuously clarified and revised through philosophy, science, law, public discussion, education, institutional reform, cultural practice, and historical experience, so that functional systems do not run for long on miscalibrated categories.

Category calibration differs from local error correction. Local error correction may address a model error, a legal loophole, a policy failure, or an educational problem. Category calibration addresses more basic questions: what counts as an error, what counts as evidence, who can bear responsibility, what constitutes judgment, what is worth optimizing, and what remains meaningful. If these questions can no longer be raised and handled, the functional systems of civilization may continue to operate, but their operation will increasingly depend on inertia, efficiency, and automation rather than understanding, judgment, and responsibility.

The philosophical kernel is therefore not a storehouse of abstract ideas, but a **dynamic mechanism** within civilizational conditions that maintains the operability of foundational categories. It exists in theoretical reflection, but also in legal procedure, scientific norms, educational training, public contestation, institutional reform, and cultural practice. Its function is not to keep civilization fixed on a single answer, but to enable civilization, when facing new technologies, new institutions, and new risks, to clarify again the basic concepts it invokes.

If the mechanism of category calibration fails, the functional systems of civilization may continue to operate while gradually losing their basis of intelligibility, accountability, and corrigibility.

4.5 The Mapping Relation among Foundational Categories, Foundational Civilizational Conditions, and Capacities of Subjecthood

To avoid conceptual drift, three related but distinct lists must be distinguished: **foundational categories**, **foundational civilizational conditions**, and **capacities of subjecthood**.

First, **foundational categories** are the conceptual interfaces that the functional systems of civilization must invoke. This paper focuses on truth, evidence, responsibility, subjecthood, value, and meaning.

Second, **foundational civilizational conditions** are the conditions under which these foundational categories remain operable within the functional systems of civilization. This paper focuses on five types:

1. the **calibrability** of truth and evidence;
2. the **attributability** of responsibility and authorization;
3. the **sustainability** of subjecthood and judgment;
4. the **openness of value-ordering to reflection**;
5. the **intelligibility and continuity** of meaning-transmission.

At the layer of foundational civilizational conditions, the six foundational categories can be summarized as five types of operating conditions: truth and evidence together constitute the calibrability of the epistemic foundation; responsibility and authorization, and subjecthood and judgment, respectively constitute the attributability and sustainability of practical reason. "Authorization" and "judgment" in the latter two pairs are not new foundational categories; they are key interfaces through which the categories of responsibility and subjecthood operate in institutional practice.

Third, **capacities of subjecthood** are the capacities actually exercised by humans under civilizational conditions. This paper focuses on six:

1. knowing what is true;
2. forming judgment;
3. bearing responsibility;
4. participating in and sustaining institutions;
5. transmitting and creating meaning;
6. acting toward the future.

These three lists are not independent parallel lists. More precisely, **foundational categories provide conceptual interfaces for foundational civilizational conditions; foundational civilizational conditions, in turn, support the actual exercise of human capacities of subjecthood**. If truth and evidence cannot be calibrated, humans struggle to know what is true and to form judgment; if responsibility and authorization cannot be attributed, humans struggle to bear responsibility and sustain institutions; if subjecthood and judgment cannot be sustained, humans struggle to participate in the common world as agents and future-oriented subjects; if value-ordering is not open to reflection, technological and institutional optimization may detach from human purposes; if meaning-transmission cannot be understood and continued, humans lose the structure of reasons for learning, creating, taking responsibility, and protecting the future.

This mapping relation also shows that this paper is not merely listing abstract concepts. It is explaining the structure of the conditions for the realization of human subjecthood.

4.6 Disambiguating the "Category of Subjecthood" and the "Realization of Subjecthood"

This paper repeatedly uses the term "subjecthood," so its two different meanings must be clarified.

As a **foundational category**, subjecthood refers to the conceptual structure concerning who can judge, consent, authorize, act, take responsibility, and represent themselves. It is part of the category-calibration mechanism. Legal systems, political systems, educational systems, ethical relations, and public governance all depend on some category of subjecthood in order to decide who can sign, who can be responsible, who can be authorized, who can consent or refuse, and who has the capacity for action and responsibility.

As part of the **realization of human subjecthood**, by contrast, subjecthood refers to the actual capacities through which humans know, judge, take responsibility, participate in institutions, create meaning, and shape the future. It is not an abstract concept, but the actual capacity through which humans unfold their mode of existence under civilizational conditions.

These two levels must not be conflated. The category of subjecthood is the categorical condition for the realization of human subjecthood; the actual realization of subjecthood is the unfolding of those categorical conditions within human civilization. The former is not a repetition of the latter, but the conceptual precondition under which the latter can be understood, institutionalized, and maintained.

This disambiguation is crucial for AI risk analysis. AI may weaken humans' actual capacities for subjecthood-realization, and it may also blur the category of subjecthood itself. The former is the weakening of capacities of subjecthood; the latter is the miscalibration of the category of subjecthood. They are related, but they must not be conflated.

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5 | Hardware–Kernel–Distribution–Application Layer: A Functional Analogy for Understanding Structural Relations

The preceding sections have explained the positions of the human species, human civilization, the philosophical kernel, and the functional systems of civilization. To place these relations more clearly within one structure, this paper introduces a limited functional analogy [Note 8]: **the human species can be analogized to hardware, the philosophical kernel to the kernel, human civilization to a distribution, and the functional systems of civilization to the application layer.**

The basic meaning of the analogy is this: the human species provides biological support; the philosophical kernel maintains foundational category interfaces; human civilization constitutes the complete intergenerational operating environment; and the functional systems of civilization are the main interfaces through which humans actually enter the common world. This paper uses the analogy not in order to understand humans as machines, philosophy as code, or civilization as software. It is only a limited functional analogy used to explain the relations of support, interface, operating environment, and application interface among the four levels.

5.1 The Limited Use and Boundary of the Analogy

This analogy serves four purposes.

First, it shows that human civilization requires embodied biological support. Just as a computing system cannot operate without physical hardware, human civilization cannot continue to unfold apart from bodies, brains, nervous systems, perceptual capacities, reproductive mechanisms, mortality, and ecological dependence. Without embodied human life, language learning, educational transmission, affective relations, responsibility-bearing, and meaning-formation lose their bearers.

Second, it shows that the functional systems of civilization require foundational category interfaces. Just as applications cannot operate apart from underlying system interfaces, systems such as science, law, education, politics, economy, culture, media, and administration cannot operate stably apart from foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. Their ability to judge, attribute, interpret, authorize, evaluate, and transmit depends on these foundational categories remaining intelligible, accountable, and corrigible.

Third, it shows that humans actually live in a complete civilizational operating environment, rather than directly in abstract categories. Ordinary people enter the common world not by directly confronting concepts such as "truth," "responsibility," or "meaning" themselves, but through families, schools, courts, markets, media, scientific communities, work organizations, technological platforms, and public institutions.

Fourth, it shows that the functional systems of civilization are the practical interfaces through which human subjecthood is realized. Through education, humans learn understanding and judgment; through science, they participate in knowledge production; through law, they understand rights and responsibility; through governance, they participate in public direction; through culture, religion, and art, they express meaning; through economy and labor, they participate in the maintenance and renewal of the common world.

The analogy helps to avoid two misunderstandings. The first is the assumption that the survival of the human species is equivalent to the intactness of the conditions for the realization of human subjecthood. The second is the assumption that because the functional systems of civilization continue to operate, the underlying categories remain healthy. In fact, a system may continue to operate on the surface, and may even operate faster, more efficiently, and more automatically, while its underlying interfaces have become unintelligible, unaccountable, or uncalibrated.

All analogies in this section are therefore subject to one limitation: they explain functional relations; they do not determine ontological relations.

5.2 Functional Explanation of the Four Layers

First, the human species as hardware. In the structural relation of human existence, the human species provides the most basic biological support. If we use a computing system as an analogy, it is like hardware. Without embodied human life, there is no human civilization, no realization of subjecthood, and no commitment to the future. Species survival therefore constitutes the baseline of existential-risk analysis: if the human species becomes extinct, or permanently loses the capacity for continuity and action, then civilizational conditions, the category-calibration mechanism, and functional systems all lose their bearer.

But hardware is not a complete operating environment. The existence of humans as a biological species does not automatically generate science, law, education, institutions, structures of responsibility, historical memory, or narratives of meaning. Species survival is a necessary condition, but not a sufficient condition.

Second, the philosophical kernel as the kernel. The philosophical kernel can be analogized, in a limited sense, to the kernel. Here "kernel" does not mean that some fixed philosophical theory governs civilization, nor that academic philosophy stands institutionally above science, law, politics, education, economy, or culture. It means the calibration mechanism within civilizational conditions that maintains the operability of foundational categories.

Science requires the interfaces of truth and evidence; law requires the interfaces of responsibility, subject, and attribution; politics requires the interfaces of legitimacy and publicity; education requires the interfaces of knowledge, growth, and purpose; economy requires the interfaces of value, preference, and the justification of distribution; culture and art require the interfaces of meaning, expression, and shared experience. The philosophical kernel does not produce all civilizational content, but it affects whether the functional systems of civilization can stably invoke these foundational categories.

If the interfaces of truth and evidence become miscalibrated, science and education may continue to produce content, but the connection between knowledge and reality becomes fragile. If the interfaces of responsibility and authorization become miscalibrated, law and administration may continue to run procedures, but the attribution between action and consequence becomes unstable. The philosophical kernel is not code, nor a module that can be separately installed, replaced, or restarted. It is called a "kernel" only because, functionally, it performs the calibration of foundational category interfaces.

Third, human civilization as a distribution. Human civilization can be analogized, in a limited sense, to a distribution. A distribution is not the hardware itself, nor the kernel itself. It organizes a complete operating environment around underlying interfaces—one that can actually be entered, maintained, and used. Correspondingly, human civilization is not foundational categories themselves, nor any single institution or technological system. It is the intergenerational common world composed of language, knowledge, law, education, economy, politics, culture, technology, structures of responsibility, historical memory, and narratives of meaning.

Human individuals enter civilization not by entering abstract categories directly, but by entering concrete environments composed of families, schools, law, media, scientific communities, work organizations, cultural traditions, technological platforms, and public institutions. As a complete operating environment, civilization enables humans to learn, judge, take responsibility, create, and transmit within a historical structure larger than individual life.

The distribution analogy also shows that an operating environment may appear complete while its underlying interfaces are miscalibrated. A society may still have schools, courts, media, markets, government, and cultural production, and these systems may even operate more efficiently. But if foundational interfaces such as truth, evidence, responsibility, subjecthood, value, and meaning become miscalibrated, the conditions for the realization of human subjecthood may still suffer structural weakening.

Fourth, the functional systems of civilization as the application layer. The functional systems of civilization can be analogized, roughly, to the application layer. Systems such as science, law, education, economy, politics, culture, media, administration, medicine, religion, and art are the main interfaces through which humans enter the common world in everyday life. The application layer handles concrete tasks: producing knowledge, adjudicating disputes, organizing power, educating the next generation, allocating resources, expressing meaning, treating illness, disseminating information, maintaining rituals, and interpreting life.

The application layer is visible, evaluable, and optimizable. Whether a system is running, whether a task is completed, whether a process is accelerated, whether users are satisfied, and whether efficiency improves are all relatively easy to observe. Modern AI also most readily displays its value at this layer: improving productivity, reducing costs, assisting writing, automating analysis, accelerating processes, personalizing services, and lowering barriers to knowledge.

But flourishing at the application layer does not equal the health of the system as a whole. A society may produce more reports, images, analyses, recommendations, evaluations, and decisions, while at the same time finding it increasingly difficult to confirm the relation between these outputs and truth, responsibility, judgment, and meaning. A functional system of civilization may continue to operate, and may even operate more efficiently; but if it gradually no longer needs human understanding, judgment, labor, creation, public participation, and responsibility-bearing, then the conditions for the realization of human subjecthood within it are weakened.

Thus, the analogy of the functional systems of civilization as the application layer provides an intermediate bridge for the AI risk analysis that follows. Humans are not merely users, consumers, or recipients of services in the functional systems of civilization. They ought also to be workers, learners, judges, creators, citizens, bearers of responsibility, and future-oriented subjects.

5.3 A Non-Analogical Statement of the Structural Relation

The analogy ends here. What has argumentative force is the following non-analogical structural account.

The human species, human civilization, the philosophical kernel, and the functional systems of civilization are not four parallel entities, nor four risk-bearing subjects. They identify different conditions under which one and the same risk-bearing subject—humanity as an embodied species—exists and realizes subjecthood.

The human species provides biological support. Without embodied human life, there is no human civilization, no realization of subjecthood, and no commitment to the future. Human civilization provides the intergenerational common world in which subjecthood is realized. Without language, knowledge, institutions, structures of responsibility, historical memory, and narratives of meaning, humans struggle to exist as knowers, judges, bearers of responsibility, institutional participants, and creators of meaning. The philosophical kernel, as a category-calibration mechanism within civilizational conditions, maintains the intelligibility, accountability, and corrigibility of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. The functional systems of civilization are the operating interfaces through which humans actually participate in these conditions, entering the common world through science, law, education, economy, culture, governance, and public services.

Therefore, AI-related existential risk to humanity cannot be understood only as the question whether a technical system harms human bodies. Nor can it be understood only as the question whether a social application causes a local problem. AI may act upon biological support; it may change the civilizational operating environment; it may weaken humans' substantive participation in functional systems; and it may disturb the calibration mechanism of foundational categories. All of these risks must ultimately return to one question: can humanity, as an embodied species, still realize its subjecthood under civilizational conditions?

The above is the non-analogical structural relation. The next section explains why the development, deployment, and embedding of AI may act upon these conditions and bring them into the scope of existential risk.

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6 | Why AI Makes These Conditions an Issue of Existential Risk

6.1 AI and the Biological-Survival Layer

First, AI may act upon the biological-survival layer of humanity. This is the most familiar part of existing discussions of AI existential risk, and it is also the baseline form of risk that this paper always preserves. If AI systems, through loss of control, misuse, competitive pressure, weaponization, high-consequence technological chains, or risks to critical infrastructure, cause the extinction of the human species, permanent incapacitation, or the irreversible foreclosure of humanity's future, then they directly constitute existential risk to humanity.

Such risk may arise through multiple pathways. Sufficiently powerful AI systems may assist dangerous biotechnology research, enhance cyberattack capabilities, affect critical infrastructures such as energy, communication, finance, transportation, and supply chains, participate in automated weapons systems, or amplify cascading consequences in complex social systems that humans cannot understand and control in time.

The importance of the biological-survival layer also lies in the fact that it is the precondition of support for all other layers and mechanisms. Without embodied human life, without bodies, nervous systems, mortality, reproductive mechanisms, and intergenerational continuity, human civilization, the philosophical kernel, structures of responsibility, and narratives of meaning cannot continue.

Moreover, civilizational conditions and the category-calibration mechanism themselves affect humanity's capacity to deal with risks at the species level. A society unable to stably distinguish truth from falsehood, or evidence from narrative, will struggle to reliably assess high-consequence technological risks. A society with unclear structures of responsibility will struggle to govern dangerous chains of action. A society whose capacities for subject judgment have weakened will struggle to form long-term public action. A society with impoverished structures of meaning may lack reasons to protect future generations. Risks at the species level and risks to the conditions for the realization of subjecthood are therefore not in competition. They are mutually dependent. The former provides the baseline; the latter explains why the human future above that baseline remains a human future.

6.2 AI as a Civilizational Mediator

The specificity of AI lies in the fact that it does not appear only as an external source of danger. It is also entering human knowledge production, institutional processes, judgment formation, responsibility allocation, and the organization of meaning as a **civilizational mediator**. It does not simply stand outside human civilization and inflict damage like a natural disaster or an external enemy. It is increasingly embedded within the functional systems of civilization, participating in their operation, acceleration, ranking, filtering, evaluation, and generation.

When AI enters educational systems, it affects not only learning efficiency, but also how humans form understanding, questioning, and judgment. When AI enters scientific and knowledge systems, it affects not only the speed of information processing, but also how evidence is organized, interpreted, and accepted. When AI enters legal and governance systems, it affects not only administrative efficiency, but also how responsibility is allocated, reasons are generated, and procedures are understood. When AI enters cultural and media systems, it affects not only content production, but also the formation of meaning, memory, shared experience, and public imagination.

The risk significance of AI therefore lies not only in the possibility that it may cause external disasters, but also in the possibility that it may change how humans know what is true, form judgment, bear responsibility, participate in institutions, and create meaning. If AI merely provides reversible, intelligible, and correctable tool support within a local function, then it remains primarily an ordinary social risk or a problem of social adaptation [Note 9]. But if AI systems begin systematically to reorganize the ways in which humans enter structures of knowledge, institutions, and meaning, and further affect the foundational conditions for subjecthood-realization, then the problem may rise to the level of existential risk at the civilizational-conditions layer.

The key question is not whether AI "participates in civilization." Writing, tools, machines, the Internet, platform systems, and automation have all participated in the operation of civilization. The truly critical question is whether AI's participation weakens the position of humans as substantive participants in the common world, as judges, bearers of responsibility, and shapers of direction.

6.3 Deprivation by Substitution

AI's impact on the civilizational-conditions layer does not appear only as a tool becoming more efficient, nor only as an industry becoming automated. The deeper problem is that AI is entering the functional systems of civilization and changing how humans participate in the common world.

As Sections 3 and 5 explained, systems such as economy, law, education, science, state governance, culture, media, administration, and public services are important support layers for the realization of human subjecthood. Humans learn, work, judge, create, take responsibility, govern, and act toward the future through these systems. As a more efficient, more replicable, and more scalable substitute, AI may gradually take over human roles in labor, cognition, creation, governance, and judgment within these systems, causing humans to be pushed out of or marginalized within them. This paper calls this mechanism **deprivation by substitution** [Note 10].

Economic systems may depend less and less on human labor and professional judgment; cultural systems may depend less and less on human creation and interpretation; governance systems may depend less and less on human public judgment and political participation; educational and research systems may increasingly depend on automated generation, automated evaluation, and automated selection. In such cases, the functional systems of civilization continue to operate, and may even operate more efficiently, but the position of humans as participants in and shapers of the direction of those systems is weakened.

This risk does not necessarily appear as humans immediately losing all rights or resources. A more common form may be **relative disempowerment**: humans still consume, vote, confirm, and receive services, but exercise decreasing real influence over resource allocation, institutional direction, knowledge production, and cultural narratives. Humans remain inside the system, but are no longer necessary participants in its operation. If this tendency crosses multiple functional systems such as economy, culture, state governance, education, and media, and is reinforced through feedback among them, it may weaken humanity's capacity, as a future-oriented subject, to shape the common world.

The core question of deprivation by substitution is this: can humans still participate **substantively** in the formation and renewal of the common world as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects? If the answer gradually becomes negative, then AI's impact on civilizational conditions is not merely a problem of social adaptation. It may touch the foundational conditions for the realization of human subjecthood.

Deprivation by substitution first erodes **substantive participation**. Humans may still retain formal participation—clicking, signing, voting, choosing, consuming—but exercise decreasing substantive influence over the direction of the common world. If such erosion crosses multiple key systems and shows an irreversible or difficult-to-reverse tendency, it may become an issue of existential risk, although it must still pass through the four thresholds specified in Section 6.6. The danger of deprivation by substitution does not lie in the disappearance of a particular job, profession, or industry, but in the possibility that humans gradually cease to be necessary subjects in the operation and renewal of the common world.

6.4 Mediated Reconfiguration

AI's impact on civilizational conditions does not only appear in the gradual replacement or exclusion of humans from the functional systems of civilization. Another, more hidden mechanism is that humans remain within the system and retain forms of choice, approval, signature, learning, creation, and participation, while the preconditions of these activities have become deeply mediated by AI. This paper calls this mechanism **mediated reconfiguration**.

Mediated reconfiguration first occurs within the functional systems of civilization. Educational systems still educate students, but learning paths, question generation, answer evaluation, and knowledge organization are increasingly mediated by AI. Legal and administrative systems still retain procedures, but evidence selection, text generation, risk ranking, and decision recommendations increasingly involve AI. Scientific and knowledge-production systems still publish research, generate explanations, and organize literature, but hypothesis generation, material selection, argumentative writing, and judgments of credibility increasingly depend on model outputs [Note 11]. Cultural and media systems still produce meaning, narrative, and public imagination, but forms of expression, the allocation of attention, and the evaluation of meaning are increasingly shaped by generative systems.

In these cases, humans have not been completely excluded from the system. They still read, confirm, vote, sign, learn, create, manage, and bear responsibility. The problem is that the structure of human participation has changed. Humans still receive information, but points of entry, summary structures, and interpretive frames may be pre-organized by systems; humans still make decisions, but visible options, risk rankings, and generated reasons may be shaped by AI [Note 12]; humans still bear responsibility, but the process by which judgment is formed becomes increasingly difficult to understand and trace; humans still learn and create, but processes of learning, standards of creation, and evaluations of meaning increasingly depend on system output.

The key difference between mediated reconfiguration and deprivation by substitution is this: deprivation by substitution gradually removes humans from their position within systems, weakening substantive participation; mediated reconfiguration preserves, and may even intensify, the formal position of humans within systems while hollowing out the substantive participation behind it. Humans still seem to judge, but the information on which judgment depends has been pre-filtered; humans still seem to choose, but the visible options have been pre-ranked; humans still seem to take responsibility, but the understanding and traceability required for responsibility have been weakened.

Mediated reconfiguration may further disturb the philosophical kernel. If truth is increasingly encountered through system-generated summaries, recommendations, and synthesized content, the distinction between truth and fabrication may drift. If evidence is increasingly selected, ranked, and interpreted by systems, the public testability of evidence may be weakened. If responsibility is increasingly assigned to nominal human approvers while the formation of judgment is deeply shaped by systems, the category of responsibility may become hollow. If subjecthood increasingly appears as choosing among system-provided options rather than understanding, questioning, and shaping the options themselves, the category of subjecthood may also be rewritten.

The deeper consequence of mediated reconfiguration is that the way the functional systems of civilization invoke foundational categories changes. This can be understood through five kinds of pressure on category interfaces. The following analysis corresponds to the foundational civilizational conditions defined in Section 4.5: the calibrability of truth and evidence, the attributability of responsibility and authorization, the sustainability of subjecthood and judgment, the openness of value-ordering to reflection, and the intelligibility and continuity of meaning-transmission. Truth and evidence together constitute the epistemic interface; responsibility and authorization, and subjecthood and judgment, respectively constitute interfaces of practical reason.

With respect to truth and evidence. Generative systems can produce fluent, coherent, and apparently credible content, but fluency is not truth, interpretability is not evidentiality, and relevance is not reliability. If humans increasingly take generated content as the starting point of knowledge, rather than sources, evidence, and processes of verification, then the calibration relation between truth and evidence becomes fragile. The problem is not merely that a given model occasionally makes mistakes. It is that the conditions under which human communities maintain verifiable knowledge structures may be changed.

With respect to responsibility and authorization. When AI participates in recommendation generation, evidence selection, risk assessment, option comparison, and the triggering of action, the fact that a human signs the final document does not necessarily mean that the human understood the process by which judgment was formed. The fact that institutions retain approval and audit procedures does not necessarily mean that they can trace how action was formed. Structures of responsibility therefore cannot rely only on formal signatures, procedural nodes, or log records. They must re-clarify the relations among action, authorization, understanding, fault, and attribution. “Authorization” here is not a new foundational category, but a key interface through which the category of responsibility operates in institutional practice.

With respect to subjecthood and judgment. Subjecthood cannot simply be equated with the final click, confirmation, or choice. If AI systems pre-organize questions, select information, generate reasons, rank options, and recommend actions, then even if humans retain the form of choice, they may increasingly participate less in the core process of judgment formation. “Judgment” here is not a new foundational category, but a core capacity through which subjecthood is exercised in practice. The question is not whether humans still use tools, but whether tools have begun to replace humans in forming reasons, understanding problems, and bearing responsibility for judgment.

With respect to value-ordering. AI often operates in the form of optimization problems, but optimization itself does not show whether the goal is legitimate, nor whether the optimized metric genuinely represents human value. A system may optimize engagement while weakening public reason; optimize productivity while damaging subjecthood; optimize convenience while lowering the capacity for judgment; optimize satisfaction while concealing the poverty of meaning. If the functional systems of civilization increasingly organize action through measurable indicators, value-ordering may be replaced by technical metrics, while reflection on value is compressed by the logic of optimization.

With respect to meaning-generation. AI can generate works, companionship, research, designs, and plans, causing humans to ask again about the foundations of creation, learning, labor, dignity, contribution, and meaning. If civilizational conditions cannot re-explain why humans should still learn, create, bear responsibility, and protect the future, then the realization of subjecthood loses an important support of meaning. The risk is not whether AI can generate valuable content, but whether humans can still understand the meaning of their own creation, responsibility, and commitment to the future under conditions in which generated content is abundant.

These pressures on interfaces are not isolated from one another. Truth requires evidence for calibration; responsibility requires subjecthood as its bearer; subjecthood requires judgment for its realization; value requires meaning for interpretation; and meaning requires truth, responsibility, and value if it is not to slide into arbitrary narrative. When AI simultaneously changes the practical conditions of multiple category interfaces, the functional systems of civilization may continue to operate, and may even operate more efficiently, but their operation increasingly depends on automation, prediction, ranking, and optimization rather than understanding, judgment, responsibility, and meaning. The problem is not that AI occasionally contaminates a judgment, but that the mutual calibration relations among truth, evidence, responsibility, subjecthood, value, and meaning are systemically rewritten. This paper calls the linked consequence of these multiple interfaces **systemic drift of the category network**.

Deprivation by substitution and mediated reconfiguration are not mutually exclusive. Humans may be gradually replaced in some functional systems of civilization, while remaining involved but deeply mediated in others. The former weakens the position of humans as participants in the common world; the latter weakens human understanding, judgment, responsibility, and meaning-creation within the process of participation. Together they point to the same question: can humanity still realize its subjecthood under civilizational conditions?

6.5 The Expanded Definition and Its Necessity

On the basis of the preceding analysis, this paper proposes the following expanded definition:

AI-related existential risk to humanity refers to the risk, arising from the development, deployment, embedding, or loss of control of AI systems, that AI may irreversibly destroy the human species; or render humanity permanently incapacitated, or otherwise permanently deprive humanity of its future; or irreversibly, or in ways

difficult to reverse, undermine the foundational conditions under which humanity, as an embodied species, realizes its subjecthood within civilization.

The three disjuncts are not mutually exclusive categories, but a structure of coverage organized by "what continues to exist." The first refers to the biological termination of the risk-bearing subject itself. The second refers to the continued existence of the risk-bearing subject while its future is permanently foreclosed. The third refers to the continued existence of the risk-bearing subject and a nominal future, while the foundational conditions for its realization of subjecthood within civilization are weakened irreversibly or in ways difficult to reverse. Entailment and overlap among the disjuncts are allowed, because the definition seeks joint coverage of the ways in which humanity may be harmed, not a formally mutually exclusive classification.

This definition preserves the core status of species-level existential risk, while extending risk analysis from a single biological-survival layer to the realization of subjecthood under civilizational conditions. It does not require all AI-induced social change to be included under existential risk, nor does it require every civilizational disturbance to be treated as existential harm. It claims only that when AI's impact touches the foundational conditions under which the realization of human subjecthood continues to hold, and when that impact has irreversible or difficult-to-reverse structural consequences, existential-risk theory cannot take biological survival alone as its object of judgment.

This kind of risk need not appear as sudden collapse. It may appear as a transfer of capacities within normal operation: humans still receive information, but verify sources less and less; still make choices, but participate less and less in defining the problem; still bear responsibility, but understand less and less about the formation of judgment; still produce content, but find it increasingly difficult to maintain the relation between truth and evidence; still have institutions, but find it increasingly difficult to trace the chain between action and consequence. At that point, society may be more efficient, more convenient, and more automated, while the conditions for the realization of human subjecthood are gradually weakened.

Accordingly, if AI existential-risk identification in practice focuses only on biological destruction pathways, it will miss a form of risk distinctive to the age of AI: the human species continues to exist, but the civilizational conditions under which it continues to exist as knowers, judges, bearers of responsibility, institutional participants, and creators of meaning are systematically weakened. The expanded definition proposed here is meant to give this form of risk a conceptual position.

6.6 The Four Thresholds for Risks to the Realization of Human Subjecthood

For a risk to enter the scope of existential risk at the level of the realization of human subjecthood, it should satisfy at least four conditions:

First, it must act upon foundational civilizational conditions, not local functions. The replacement of a profession, the restructuring of an industry, or the automation of an institutional process does not automatically constitute existential risk. Only when such changes systematically affect the calibrability of truth and evidence, the attributability of responsibility and authorization, the sustainability of subjecthood and judgment, the openness of value-ordering to reflection, and the intelligibility and continuity of meaning-transmission can they enter the scope of existential risk at the level of the realization of human subjecthood.

Second, it must have diffusion capacity across domains and civilizational interfaces. [Note 13] Local errors, single accidents, isolated misuses, and even automation-driven reorganization within a single state or a single institutional system do not normally constitute existential risk to humanity. Only when a mechanism can pass through global research networks, educational systems, digital platforms, legal-governance interfaces, supply chains, and structures of public discourse, cross multiple functional systems of civilization, and repeatedly change human judgment, responsibility, participation, or meaning structures does it have the risk profile of subjecthood-realization at the scale of humanity.

Third, it must be capable of causing irreversible or difficult-to-reverse structural damage. [Note 14] Existential risk is concerned not only with scale, but also with reversibility. At the civilizational-conditions layer, irreversible or difficult-to-reverse primarily means difficult to recover in historical, institutional, and cognitive terms. For example, long-term institutional path dependence may make certain automated processes default structures; ruptures in intergenerational capacity formation may cause later generations no longer to possess certain capacities for judgment, verification, and responsibility-bearing; deep reshaping of cognitive habits may cause people no longer to recognize the loss of certain capacities as a loss. This **difficulty of reversal at the civilizational-conditions layer** differs from the physical irrecoverability of species extinction, but at historical scale it may be sufficient to change humanity's capacity to continue as a future-oriented subject. Here, "difficult to reverse" does not mean high governance cost, long recovery time, or difficult institutional reform. It means that, on an intergenerational historical scale, relevant capacities, institutional pathways, and category-calibration mechanisms have deteriorated to the point where realistic agents of repair and realistic pathways of repair are lacking.

Fourth, it must cause the harms identified in the first three thresholds to converge into overall, intergenerational damage to humanity's capacity as a future-oriented subject. Existential risk is not just any major loss; it is risk that affects whether humanity can continue as the subject of its own future. The fourth threshold does not repeat the first threshold. It asks whether the first three thresholds jointly converge into overall, intergenerational damage to humanity's future-oriented capacities: if a mode of AI embedding not only acts upon foundational civilizational conditions, has diffusion capacity across domains and civilizational interfaces, and causes irreversible or difficult-to-reverse structural damage, but also makes it increasingly difficult for humanity to understand risks, revise institutions, organize action, and shape the future, then it is not merely a problem of social adaptation. It is a problem in which the foundational conditions for the realization of human subjecthood are damaged.

"Humanity as a future-oriented subject" does not introduce a new risk-bearing subject, nor is it a concept independent of subjecthood-realization. It refers to the temporal and agential dimension of subjecthood-realization: humanity not only knows, judges, and takes responsibility in the present, but also, across generations, can understand risks, revise institutions, organize action, and shape the future. The fourth threshold is therefore an aggregative criterion after the first three thresholds. It asks whether local damage to capacities has converged into damage to the overall future-directedness of humanity.

Together, these four thresholds limit the scope of the expanded definition. They ensure that this paper does not elevate all AI-related social impacts into existential risks, and does not misread ordinary technological change as civilizational-level harm. AI-related social impacts that fail to meet these thresholds are called ordinary social risks in this paper: they may act upon local functions, institutional processes, or industry structures; they may be important risks with relatively determinate paths of repair; but they are not necessarily existential risks.

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7 | Objections, Boundaries, and Paradigm Extension

The preceding sections have argued that AI-related existential risk to humanity should not be understood only as species extinction, permanent incapacitation, or future deprivation. It should also include the irreversible or difficult-to-reverse weakening of the foundational conditions under which humanity, as an embodied species, realizes subjecthood under civilizational conditions. This section responds to objections through **test cases** and **engagement with adjacent literature**, and clarifies the boundary of the paradigm extension.

7.1 Testing the Thresholds: Two Cases

Objection 1: The paper may overinflate ordinary social problems such as job replacement, information pollution, educational dependence, or institutional automation into existential risks.

This paper explicitly rejects such overextension. AI-related ordinary social risk is not the same as AI-related existential risk to humanity. Ordinary social risk may be very serious, may cause widespread harm, and may require strong governance responses. But it usually acts upon a local social function, institutional process, industry structure, or technological application, and has a relatively determinate governance object and path of repair.

The following two cases illustrate the filtering function of the four thresholds.

Case A (excluded): Suppose that 80 percent of manufacturing jobs in a country are replaced by AI automation, but the educational system continues to train a new generation of engineers, the legal system remains led by human judges in evidence review and responsibility attribution, and the scientific community continues independently to propose hypotheses, verify evidence, and trace errors. Manufacturing workers can enter other fields through retraining, and institutional direction continues to be shaped by human public judgment. This is an **ordinary social risk**: it primarily acts upon a local function—the labor structure of the economic system—has not yet acted upon foundational civilizational conditions, does not have diffusion capacity across domains and civilizational interfaces, and can be reversed through educational adjustment, institutional repair, and intergenerational retraining. It does not pass the four thresholds.

Case B (included): Suppose that, over the next twenty years, the global scientific publishing system comes to rely comprehensively on AI-generated literature reviews, experimental designs, statistical-significance screening, and peer-review assistance, causing a new generation of researchers no longer to possess the capacity independently to propose hypotheses, verify evidence, and trace errors. At the same time, legal systems comprehensively adopt AI-generated judicial reasoning and risk ranking, causing "responsibility attribution" to degenerate into formal signature. Educational systems use AI-personalized learning to deeply organize learning paths, question generation, and evaluation standards, causing public discussion, critical questioning, and cross-subjective argumentative training gradually to recede. This case primarily manifests **mediated reconfiguration** rather than deprivation by substitution: humans retain the formal positions of scientist, judge, and learner, but the preconditions of judgment formation, evidence testing, and the understanding of meaning have been deeply reorganized by AI.

This case **acts upon foundational civilizational conditions**—the calibrability of truth and evidence, the **attributability of responsibility and authorization**, the sustainability of subjecthood and judgment, the **openness of value-ordering to reflection**, and the **intelligibility and continuity of meaning-transmission**. It involves **diffusion across domains and civilizational interfaces**: global scientific publishing, legal-governance interfaces, and educational training are simultaneously reconfigured, and this reconfiguration spreads across regions through digital platforms, research networks, and institutional standards. It is **difficult to reverse**: ruptures in intergenerational capacity formation make independent research, critical judgment, and responsibility attribution scarce capacities. It also **converges into overall damage to future-oriented subject capacities**: humanity can no longer independently revise knowledge, bear responsibility, and act toward the future. It therefore passes the four thresholds and enters the scope of existential risk.

The contrast between these two cases shows that the four thresholds are not a decorative list, but an operational filtering mechanism. They ensure that this paper does not elevate all AI-related social impacts into existential risks, and also does not reduce existential risk to species extinction alone.

7.2 Are Civilization or the Philosophical Kernel Being Reified?

Objection 2: In discussing civilizational conditions, the philosophical kernel, and category-calibration mechanisms, the paper seems to turn civilization or the philosophical kernel into a new risk-bearing subject.

This misunderstanding must be rejected. The core claim of the paper is precisely that the risk-bearing subject of AI-related existential risk to humanity is always humanity as an embodied species. Civilization, institutions, cultures, technological systems, and the philosophical kernel are not risk-bearing subjects alongside humanity. Civilizational conditions matter not because civilization itself becomes an independent subject, but because humanity as an embodied species needs civilizational conditions in order to realize subjecthood. The philosophical kernel is likewise not an entity. It is not academic philosophy, not the highest department of some civilizational system, and not a mysterious principle hidden behind civilization. It refers to the category-calibration mechanism within civilizational conditions that maintains the operability of foundational categories.

Any interpretation that treats civilization, institutions, or the philosophical kernel as independent risk-bearing subjects departs from the core framework of this paper.

7.3 Is This Merely Another Name for AI Disempowerment or Autonomy Degradation?

Objection 3: The paper may seem simply to redescribe existing discussions in new language, such as AI disempowerment, autonomy degradation, over-reliance, deskilling, or the decline of human agency.

This paper acknowledges that these discussions importantly overlap with its concerns. AI may weaken human capacities, increase dependence, transfer power, change decision structures, and reduce the training of human judgment. These are important clues for understanding AI risk.

But the task of this paper is not simply to repeat these discussions. It is to return them to a more prior framework of risk identification. Declines in capacity, over-reliance, transfers of power, or reductions in autonomy usually describe changes in particular capacities, institutional relations, or positions of action. This paper asks a more basic question: under what conditions do these changes cease to be merely local capacity degradation and instead touch existential harm to humanity as the risk-bearing subject?

More specifically, the “gradual disempowerment” described by Kulveit et al. (2025) is adjacent, at the level of phenomena, to what this paper calls “deprivation by substitution.” But the former is located in the social dynamics of power transfer, whereas this paper asks: under what conditions does such transfer act upon foundational civilizational conditions, have diffusion capacity across domains and civilizational interfaces, become difficult to reverse, and converge into overall damage to future-oriented subject capacities? The analyses by Prunkl (2022) and Krook (2025) of autonomy decline and skill degradation [Note 15] intersect, at the level of symptoms, with this paper’s “mediated reconfiguration.” But this paper further asks: when autonomy decline is not the result of individual choice, but a systemic drift of the category-calibration mechanism within the functional systems of civilization, does it touch the foundational conditions for the realization of human subjecthood?

The difference between this paper and these discussions therefore lies not in denying them, but in changing the level of analysis. This paper does not begin from a particular capacity or institutional relation. It begins from the risk-bearing subject and its layers of possible harm. Within this framework, disempowerment, autonomy degradation, over-reliance, and deskilling can be repositioned as possible mechanisms or symptoms, not as the final conceptual center of the paper. The conceptual center of this paper is not any single decline in capacity, but the question of when such declines rise to the level of existential harm to humanity as the risk-bearing subject.

8 | Conclusion: Before Protecting Humanity, Clarify What Humanity Is

This paper began from a prior question: when we say that AI may threaten "humanity," what exactly is the "humanity" at risk?

The paper has constructed an exercise in **conceptual engineering** that distinguishes two basic layers of possible harm to the risk-bearing subject: the biological-survival layer and the civilizational-conditions layer. Within the civilizational-conditions layer, it further identifies the philosophical kernel: the category-calibration mechanism of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. The biological-survival layer explains whether humanity continues to exist as an embodied species; the civilizational-conditions layer explains whether humanity can still realize subjecthood through an intergenerational common world; the philosophical kernel explains whether these foundational categories can still be clarified, maintained, and adjusted.

The risk-bearing subject is always humanity as an embodied species. This paper does not extend the risk-bearing subject to civilization, institutions, the philosophical kernel, or AI systems. It extends the ways in which one and the same subject may suffer existential harm: from destruction, permanent incapacitation, or future deprivation at the biological-survival layer alone, to the irreversible or difficult-to-reverse weakening of the foundational conditions under which humanity realizes its subjecthood under civilizational conditions.

The paper has analyzed two mechanisms through which AI acts upon the civilizational-conditions layer: **deprivation by substitution** and **mediated reconfiguration**. Deprivation by substitution gradually removes humans from substantive participation in the functional systems of civilization, leaving formal participation while hollowing out directional influence. Mediated reconfiguration preserves the formal position of humans, but deeply pre-organizes information entry, reason generation, option ranking, and responsibility tracing through AI. The two mechanisms point to the same question: can humanity still realize its subjecthood under civilizational conditions?

Accordingly, this paper understands AI-related existential risk to humanity as a structure of joint coverage among three modes of harm: the biological termination of the human species, the permanent foreclosure of humanity's future, and the irreversible or difficult-to-reverse weakening of the foundational conditions under which humanity realizes subjecthood within civilization.

What the paper provides is not a complete taxonomy of risks, not a technical-control scheme, and not a policy checklist, but the **ontological coordinate system** needed at the level of risk identification in AI safety, risk analysis, and governance. This coordinate system enables different lines of research to answer a more prior question before discussing specific technologies or policies: what exactly are we trying to protect?

Before protecting humanity, we must first clarify what humanity is. Before governing AI risk, we must first explain which kinds of harm truly touch the foundational conditions under which humanity can continue to exist as a future-oriented subject. **Otherwise, humanity may find itself in a civilization that operates more efficiently while losing the foundations of its existence as the subject of its own future.**

In this sense, the key question of existential risk is not whether AI changes society, but whether that change makes humanity increasingly unable to be the author of its own future.

Appendix A | Core Conceptual Glossary

This appendix gathers and defines the core concepts used in this paper. Its purpose is not to restate the argument of the main text, but to provide stable terminological boundaries for the conceptual engineering carried out in the paper. Each entry specifies, as far as necessary, the concept's definition, boundaries, and role in the paper's argument. Unless otherwise indicated, all concepts in this appendix serve the paper's analysis of the risk-bearing subject of AI-related existential risk to humanity.

I. Core Ontological Concepts

A.1 Conceptual Engineering

Conceptual engineering refers to the work of making explicit, evaluating, revising, and reconstructing the use, boundary conditions, implicit assumptions, and theoretical function of a concept. It does not merely explain the ordinary meaning of a concept in existing language. Rather, it asks whether the concept is sufficiently clear, whether it is misleading, and whether it needs to be redrawn within a particular theoretical and practical context.

In this paper, the object of conceptual engineering is "humanity" as the risk-bearing subject of AI-related existential risk to humanity. The paper does not attempt to redefine all AI risks, nor does it directly propose a governance scheme. Instead, it asks: when we say that AI may threaten "humanity," must humanity be understood only as a biological species, or must the analysis also include the foundational conditions under which humanity, as an embodied species, realizes its subjecthood under civilizational conditions?

The conceptual-engineering function of this paper is to transform "humanity" from an unexamined background term into a core analytic object in risk identification.

A.2 AI-Related Existential Risk to Humanity

AI-related existential risk to humanity refers to the risk, arising from the development, deployment, embedding, or loss of control of AI systems, that AI may irreversibly destroy the human species; or render humanity permanently incapacitated, or otherwise permanently deprive humanity of its future; or irreversibly, or in ways difficult to reverse, undermine the foundational conditions under which humanity, as an embodied species, realizes its subjecthood within civilization.

This definition includes three modes of harm. First, humanity as a biological species may be irreversibly destroyed. Second, humanity may survive without becoming extinct, yet become permanently incapacitated or otherwise permanently lose an open future. Third, the human species and a nominal future may continue to exist, while the foundational conditions for the realization of human subjecthood under civilizational conditions are irreversibly, or in ways difficult to reverse, weakened.

These three modes of harm are not mutually exclusive categories, but a structure of coverage. Species extinction normally also entails future deprivation; permanent incapacitation is one core form of future deprivation; and the irreversible weakening of the conditions for the realization of subjecthood shows that even if humans continue to exist biologically, humanity may lose the conditions under which it can continue to exist as a future-oriented subject.

A.3 Risk-Bearing Subject

A **risk-bearing subject** is the entity that, within a given risk analysis, is regarded as liable to suffer fundamental, irreversible, or difficult-to-reverse harm. It answers the question: upon whom, or upon what, does the risk ultimately fall?

In this paper, the risk-bearing subject of AI-related existential risk to humanity is always humanity as an embodied species. Civilization, states, cultures, institutions, technological systems, the philosophical kernel, and AI systems may all function as structures through which risks arise, propagate, intensify, or become mediated. They are not, however, the risk-bearing subject in the sense used in this paper.

This concept prevents a common confusion: mistaking the system in which a risk occurs for the subject that bears the risk. AI may act upon economic systems, legal institutions, educational structures, knowledge systems, or category-calibration mechanisms. These effects become existentially significant only because they may damage the biological survival of humanity as an embodied species, or the foundational conditions under which humanity realizes its subjecthood within civilization.

A.4 Humanity

In this paper, **humanity** refers first to *Homo sapiens* as an embodied biological population. Humanity has bodily structures, nervous systems, capacities for perception, reproductive mechanisms, physiological vulnerability, mortality, and dependence on ecological and material environments.

Humanity, however, does not exist only at the level of biological survival. Humanity as an embodied species also realizes subjecthood under civilizational conditions: through language, knowledge, education, institutions, structures of responsibility, public reason, historical memory, narratives of meaning, and functional systems of civilization, humans acquire and sustain the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future.

Thus, "humanity" in this paper has two basic analytic layers: the biological-survival layer and the layer of subjecthood-realization under civilizational conditions. The former identifies the embodied bearer of human existence; the latter explains why humanity is not merely a biological population, but a historical form of existence capable of being a knower, judge, bearer of responsibility, participant in institutions, creator of meaning, and future-oriented subject.

A.5 Embodied Species

An **embodied species** means that humanity is not a pure information structure, an abstract rational subject, or a computational pattern that can be arbitrarily migrated, but a biological population constituted by bodies, nervous systems, perception, affect, vulnerability, mortality, relations of dependence, and intergenerational continuity.

Embodiment is not an external attribute added to human subjecthood. It is the biological precondition under which human civilization, responsibility, meaning, and future commitment can be borne. Human judgment, learning, responsibility-bearing, and meaning-formation are tied to the body, finitude, temporality, and the capacity to live with the consequences of action.

In AI risk analysis, the concept of the embodied species resists a reductive tendency: the tendency to understand humans as cognitive patterns replaceable by more efficient information-processing systems. This paper argues that even if AI can perform many cognitive tasks, humans cannot be reduced to information-processing structures, because human existence as responsible agents, actors, and meaning-creators depends on embodied finitude.

A.6 Biological-Survival Layer

The **biological-survival layer** refers to the survival of humanity as *Homo sapiens*, an embodied biological population. It includes bodily structure, nervous systems, perceptual capacities, reproductive capacity, physiological vulnerability, mortality, and ecological dependence.

The biological-survival layer constitutes the physical bearer of human existence. If the human species becomes extinct, or if humanity permanently loses the capacity to reproduce, act, or shape the future, then human civilization, structures of responsibility, narratives of meaning, and the realization of subjecthood all lose their bearer.

In this paper, the biological-survival layer corresponds to the baseline forms of traditional existential risk, including species extinction, permanent incapacitation, and the permanent foreclosure of humanity's future. The paper's expanded account of existential risk does not weaken the centrality of the biological-survival layer; rather, it builds upon that foundation in order to analyze modes of harm at the civilizational-conditions layer.

A.7 Civilizational-Conditions Layer

The **civilizational-conditions layer** refers to the structural conditions upon which humanity, as an embodied species, depends in order to realize subjecthood within civilization. These conditions include language, knowledge systems, educational structures, legal institutions, mechanisms of responsibility, public reason, scientific communities, historical memory, technological practices, narratives of meaning, and functional systems of civilization.

The civilizational-conditions layer is not a subject parallel to humanity, nor is it a risk-bearer independent of humanity. It is a constitutive condition for the realization of human subjecthood. If these conditions are weakened irreversibly or in ways difficult to reverse, the subject harmed is not "civilization itself" as an independent subject, but humanity as an embodied species in its capacity to realize subjecthood within civilization.

This concept shows that existential risk to humanity need not appear only as bodily destruction or species extinction. It may also appear as a condition in which humans remain alive, yet gradually lose the conditions for knowing what is true, forming judgment, bearing responsibility, participating in institutions, transmitting meaning, and acting toward the future.

A.8 Human Civilization

Human civilization is the intergenerational common world formed by humanity, as an embodied species, through language, knowledge, institutions, law, education, structures of responsibility, technological practices, functional systems of civilization, and narratives of meaning.

Human civilization is not the sum of population, is not identical with the state, is not merely culture, and cannot be reduced to a technological system. Population provides biological support; the state provides political and legal organization; culture provides meaning-expression and a lived interface; technology provides capacities for action. Human civilization is the common world formed through the intergenerational transmission, institutional organization, accumulation of knowledge, and generation of meaning among these elements.

This paper does not offer a general theory of civilization. Unless otherwise specified, "civilization" in the main text refers to human civilization. Human civilization is "human" not because it contains no nonhuman technological components, but because its risk-bearing subject is humanity as an embodied species and because its normative focus is the realization of human subjecthood.

A.9 Intergenerational Common World

An **intergenerational common world** is the common world that, through language, memory, education, institutions, knowledge, structures of responsibility, and narratives of meaning, enables the past to be transmitted to the future and enables individual lives to enter a historical and interpretive space larger than themselves.

An intergenerational common world is not the simple continuation of a population through time, nor is it the mechanical preservation of cultural materials. It requires that later generations not only be born biologically, but also enter a common world that can be understood, inherited, criticized, revised, and renewed.

In this paper, the intergenerational common world is a core defining feature of human civilization. It distinguishes "human civilization" from population size, state organization, cultural custom, and technological systems, and it also provides the temporal basis for "humanity as a future-oriented subject." The true continuity of civilizational conditions is not merely that "some humans remain alive," but that "some humans remain capable of understanding, inheriting, criticizing, and renewing a common world."

A.10 Modern Global Civilizational Ecology

The **modern global civilizational ecology** refers to the highly interconnected civilizational environment of contemporary human society, formed by multiple civilizational traditions, scientific systems, technological infrastructures, legal institutions, markets, states, international organizations, digital platforms, educational systems, media systems, and public reason.

The concept emphasizes two points. First, human civilization as a whole is not the universalization of one civilizational tradition, nor the enlargement of one state or culture. Second, modern global civilization has developed a number of shared interfaces, such as scientific systems, international law, public-health systems, Internet protocols, financial networks, digital platforms, and AI infrastructures.

In this paper, the concept of modern global civilizational ecology explains why AI risks do not act only upon local states, cultures, or platforms. They may also act upon the knowledge, institutional, technological, and meaning-bearing interfaces upon which multiple civilizational traditions jointly depend. It is also an important background for determining whether a risk at the civilizational-conditions layer reaches the scale of humanity as such.

II. Concepts of Civilizational Conditions and the Realization of Subjecthood

A.11 Constitutive Condition

A **constitutive condition** is a necessary condition under which a capacity, identity, or mode of existence can come into being. It is not an external or replaceable auxiliary tool.

Instrumental conditions help a subject perform a task; constitutive conditions participate in forming the subject's very capacity to perform that task. Language, for example, is not only a tool for expressing thought, but also a condition under which complex thought, the exchange of reasons, and normative commitment can unfold. Similarly, education, structures of responsibility, public reason, functional systems of civilization, and narratives of meaning are not merely external supports for human subjecthood; they are constitutive conditions under which humans can exist as knowers, judges, bearers of responsibility, institutional participants, and creators of meaning.

This paper uses the concept to show that civilizational conditions are not decorative additions to human subjecthood, but part of the realization of human subjecthood itself. Once civilizational conditions are structurally weakened, what is damaged is not merely an external environment, but the conditions under which the realization of human subjecthood can come into being.

A.12 Functional Systems of Civilization

Functional systems of civilization are the systems within human civilization that concretely bear knowledge production, institutional action, allocation of responsibility, educational transmission, economic organization, cultural expression, public governance, medical care, media communication, and the creation of meaning.

Typical functional systems of civilization include economic systems, state and governance systems, legal systems, educational systems, scientific and knowledge-production systems, cultural and media systems, medical systems, administrative systems, infrastructure systems, religious systems, and artistic systems.

Functional systems of civilization form an intermediate support layer between abstract civilizational conditions and concrete social mechanisms. Humans usually do not encounter "civilization" as an abstract whole. They enter civilizational conditions through schools, courts, laboratories, markets, media, government agencies, platform systems, families, religious traditions, artistic practices, and public services.

Functional systems of civilization are not the risk-bearing subject, but they are a key channel through which AI acts upon the conditions for the realization of human subjecthood. AI's effects at the civilizational-conditions layer often first appear as a reorganization of human participation within these functional systems.

A.13 Structures of Human Participation

Structures of human participation are the ways in which humans participate in the operation, maintenance, and renewal of the common world as workers, learners, judges, creators, citizens, bearers of responsibility, and future-oriented subjects within the functional systems of civilization.

Human participation is not mere formal presence. It requires that humans be able to understand processes, form judgment, bear responsibility, create meaning, and exert substantive influence on institutional direction and the public future. If humans retain only formal confirmation within functional systems, while no longer genuinely understanding, judging, taking responsibility, creating, or shaping direction, then the realization of human subjecthood under civilizational conditions is weakened.

This concept provides the basis for "deprivation by substitution" and "mediated reconfiguration." Deprivation by substitution weakens the human position within functional systems; mediated reconfiguration preserves the form of human participation while altering the preconditions of judgment, understanding, responsibility, and meaning-creation.

A.14 Formal Participation and Substantive Participation

Formal participation and substantive participation are a pair of concepts used to distinguish different states of human participation within the functional systems of civilization.

Formal participation refers to a mode of participation in which humans retain visible positions, procedural actions, and nominal roles within functional systems of civilization, such as clicking to confirm, signing documents, approving recommendations, choosing options, voting, consuming, and giving feedback. Formal participation preserves the recognizability of humans within institutions, but it does not guarantee that humans genuinely understand, judge, take responsibility, or shape the process.

Substantive participation refers to a mode of participation in which humans, as knowers, judges, bearers of responsibility, maintainers of institutions, creators of meaning, and future-oriented subjects, exert real influence on the operation, direction, and renewal of the common world. It requires that humans be able to understand processes, form judgment, bear responsibility, create meaning, and exert substantive influence on institutional direction and the public future.

This distinction is a key interface in the paper's analysis of AI risk. Deprivation by substitution typically erodes substantive participation, making humans progressively less necessary to the operation and renewal of the common world; mediated reconfiguration may preserve or even intensify formal participation while hollowing out the understanding, judgment, responsibility, and meaning-creation behind it.

A.15 Realization of Subjecthood / Realization of Human Subjecthood within Civilization

Realization of subjecthood refers to the process and condition in which humanity, as an embodied species, actually acquires and sustains the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future under civilizational conditions.

It is neither a merely biological attribute of the human species nor an independent attribute of civilization itself. It is the actual unfolding of human subjecthood under civilizational conditions. It is precisely because humans can enter structures of knowledge, responsibility, institutions, meaning, and future-directed action within civilization that humanity is not merely a biological population.

In this paper, the realization of subjecthood is the central object of risk at the civilizational-conditions layer. If AI systematically weakens these capacities and produces irreversible or difficult-to-reverse structural consequences, it may constitute an existential risk at the level of the realization of human subjecthood.

A.16 Capacities of Subjecthood

Capacities of subjecthood are the set of capacities actually exercised by humans under civilizational conditions as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects.

This paper focuses on six capacities of subjecthood:

1. knowing what is true;
2. forming judgment;
3. bearing responsibility;
4. participating in and sustaining institutions;
5. transmitting and creating meaning;
6. acting toward the future.

These six capacities are not isolated from one another. Knowing what is true supports judgment; judgment supports responsibility; responsibility and institutional participation are mutually dependent; and meaning provides the justificatory structure for learning, creation, and commitment to the future. The systemic weakening of these capacities is the central object of this paper's expanded analysis of existential risk.

A.17 Foundational Civilizational Conditions

Foundational civilizational conditions are the minimal operating conditions required for humans to realize subjecthood within civilization. They are not the whole content of human civilization, but the foundational interfaces that allow humanity to continue existing as a future-oriented subject.

This paper focuses on five types of foundational civilizational conditions:

1. the calibrability of truth and evidence;
2. the attributability of responsibility and authorization;
3. the sustainability of subjecthood and judgment;
4. the openness of value-ordering to reflection;
5. the intelligibility and continuity of meaning-transmission.

These conditions are used to determine whether a risk at the civilizational-conditions layer touches the foundations of subjecthood-realization. If these conditions are systematically weakened, humans may still live in a highly functioning society, yet become increasingly unable to know what is true, form judgment, bear responsibility, participate in institutions, transmit meaning, and shape the future.

III. Concepts of Category Calibration

A.18 Foundational Categories

Foundational categories are the basic conceptual interfaces that the functional systems of civilization must share and invoke. They enable civilization to distinguish truth from falsehood, evidence from narrative, responsibility from exemption, subject from tool, value from metric, and meaning from arbitrary expression.

This paper focuses on six foundational categories:

1. truth;
2. evidence;
3. responsibility;
4. subjecthood;
5. value;
6. meaning.

Foundational categories are not theoretical terms monopolized by any one civilizational tradition, nor do they claim to exhaust all possible core categories of civilization. This paper selects these six foundational categories because they constitute the minimal conditions for AI risk identification itself: without truth and evidence, risk cannot be known; without responsibility and subjecthood, action cannot be attributed; without value and meaning, risk governance cannot state what it is protecting or why the future matters.

Different civilizational traditions may understand these categories differently. Yet insofar as a civilizational system requires knowledge, institutions, responsibility, action, and meaning, it must invoke these categories in some form.

A.19 Philosophical Kernel

The **philosophical kernel** is the category-calibration mechanism, internal to civilizational conditions, that maintains the operability of foundational categories. It is not the collection of foundational categories themselves, but the mechanism through which foundational categories are made explicit, tested, clarified, criticized, maintained, and revised.

The philosophical kernel has the character of a meta-mechanism. Here “meta” does not mean that it constitutes a higher entity outside civilizational conditions, nor that it becomes a new risk-bearing subject. It means that what it calibrates is not a local rule internal to one functional system of civilization, but the foundational category interfaces jointly invoked by functional systems such as science, law, education, politics, economy, and culture.

The philosophical kernel is not academic philosophy as a discipline, not an independent entity within civilization, and not a risk-bearing subject alongside human civilization. Nor is it a module that can be separately installed, replaced, or restarted. It is called a "kernel" because it functionally performs the calibration of foundational category interfaces.

In this paper, the philosophical kernel explains how the functional systems of civilization can continue to operate while losing their intelligibility, accountability, and corrigibility when their underlying categories become miscalibrated.

A.20 Category-Calibration Mechanism

The **category-calibration mechanism** is the historical process through which foundational categories are made explicit, tested, criticized, maintained, and revised. It is not the function of any single institution, but a continuing process distributed across philosophy, science, law, education, public discourse, institutional reform, cultural practice, and historical experience.

The category-calibration mechanism concerns questions such as: what counts as truth, what counts as evidence, who can bear responsibility, what constitutes a subject, what is worth optimizing, and what remains meaningful.

When the category-calibration mechanism fails, the real-world manifestation is that systems such as science, law, education, governance, media, and culture may continue operating, yet become increasingly unable to stably distinguish truth from fabrication, evidence from narrative, responsibility from exemption, subject judgment from formal confirmation, value goals from optimization metrics, and meaning-creation from content generation.

A.21 Category of Subjecthood

The **category of subjecthood** is the conceptual structure concerning who can judge, consent, refuse, authorize, act, bear responsibility, and represent themselves. It is one of the foundational categories and serves functional systems of civilization such as law, politics, education, ethics, and public governance.

The category of subjecthood is distinct from the realization of subjecthood. The category of subjecthood is the conceptual interface through which the capacities of subjects are understood and institutionalized; the realization of subjecthood is the process by which humans actually exercise capacities such as knowing what is true, forming judgment, bearing responsibility, participating in institutions, creating meaning, and shaping the future.

This distinction avoids circularity. One condition for the realization of subjecthood is the clarity of the category of subjecthood; the claim is not the tautology that "subjecthood depends on subjecthood." AI may weaken humans' actual capacities for subjecthood-realization, and it may also blur the category of subjecthood itself. The two are related, but they must not be conflated.

A.22 Category Network

The **category network** is the mutually supporting relation among foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning.

Truth requires evidence for calibration; responsibility requires subjecthood as its bearer; subjecthood requires judgment for its realization; value requires meaning for interpretation; and meaning requires truth, responsibility, and value if it is not to collapse into arbitrary narrative. The miscalibration of any one foundational category may affect the understanding and operation of the others.

The concept of the category network explains why AI's impact on foundational categories is often not a single-point disturbance, but may instead appear as a linked drift among multiple category interfaces.

A.23 Systemic Drift of the Category Network

Systemic drift of the category network refers to the linked miscalibration among truth, evidence, responsibility, subjecthood, value, and meaning when AI or other structural forces simultaneously alter the practical conditions under which multiple category interfaces operate.

In such a condition, the functional systems of civilization may continue operating, and may even operate more efficiently, but their operation increasingly depends on automation, prediction, ranking, and optimization rather than understanding, judgment, responsibility, and meaning.

This concept describes the deeper consequence of mediated reconfiguration. The risk is not merely that a given system malfunctions, nor merely that a model contaminates a particular judgment, but that the way functional systems of civilization invoke foundational categories undergoes systemic drift.

A.24 Mapping Relation among Foundational Categories, Foundational Civilizational Conditions, and Capacities of Subjecthood

The **mapping relation among foundational categories, foundational civilizational conditions, and capacities of subjecthood** refers to the functional connection among three levels in this paper:

1. foundational categories provide conceptual interfaces;
2. foundational civilizational conditions provide operating conditions;
3. capacities of subjecthood are the capacities actually exercised by humans within civilization.

This mapping relation can be summarized as follows: foundational categories support foundational civilizational conditions, and foundational civilizational conditions support capacities of subjecthood.

The concept prevents the three lists from drifting apart. It shows that the paper is not arbitrarily listing abstract concepts, but constructing a layered relation from conceptual interfaces to civilizational operating conditions and then to actual human capacities.

IV. Concepts of AI Risk Mechanisms and Thresholds

A.25 AI as a Civilizational Mediator

AI as a civilizational mediator means that AI is not merely an external tool, but may enter the structures of knowledge, evidence, responsibility, judgment, institutions, and meaning within human civilization, becoming a mediating force that affects the conditions for the realization of human subjecthood.

AI may become a tool, platform, infrastructure, or mediating structure within human civilization, but it does not thereby become the risk-bearing subject in this paper. The risk significance of AI lies not only in the possibility that it may cause external disasters, but also in the possibility that it may change how humans know what is true, form judgment, bear responsibility, participate in institutions, and create meaning.

This concept explains the specificity of AI risk: AI does not only stand outside human civilization and inflict harm; it may also become embedded inside the functional systems of civilization and change the way humans enter structures of knowledge, institutions, responsibility, and meaning.

A.26 Deprivation by Substitution

Deprivation by substitution refers to the process by which AI, as a more efficient, more replicable, and more scalable substitute, progressively takes over human roles in labor, cognition, creation, governance, and judgment within the functional systems of civilization, thereby marginalizing humans within, or displacing them from, the key functional systems of the common world.

Deprivation by substitution primarily weakens the position of humans as participants in, and shapers of the direction of, the common world. Its typical form is not the immediate disappearance of humans, but a condition in which humans still consume, confirm, vote, or receive services, while exercising decreasing substantive influence over resource allocation, institutional direction, knowledge production, and cultural narratives.

The danger of deprivation by substitution does not lie in the disappearance of a particular job, profession, or industry, but in the possibility that humans gradually cease to be necessary subjects in the operation and renewal of the common world. This concept corresponds to damage to structures of human participation and substantive participation at the civilizational-conditions layer.

A.27 Mediated Reconfiguration

Mediated reconfiguration refers to the condition in which humans remain within the functional systems of civilization and retain forms of choice, approval, signature, learning, creation, and participation, while the preconditions of these activities have become deeply mediated by AI.

In mediated reconfiguration, humans still receive information, but the points of entry, summary structures, and interpretive frames of that information may be pre-organized by systems; humans still make decisions, but the visible options, risk rankings, and generated reasons may be shaped by AI; humans still bear responsibility, but the process by which judgment is formed becomes increasingly difficult to understand and trace; humans still learn and create, but learning processes, standards of creation, and evaluations of meaning increasingly depend on system output.

Mediated reconfiguration primarily weakens human understanding, judgment, responsibility, and meaning-creation within the process of participation, and may further disturb the mechanism of category calibration. It differs from deprivation by substitution: deprivation by substitution progressively removes humans from key positions within systems; mediated reconfiguration preserves the formal position of humans while altering the judgment, responsibility, and meaning structures behind those positions.

A.28 Irreversible or Difficult-to-Reverse

Irreversible or difficult-to-reverse is the key temporal standard used in this paper to distinguish existential risk from ordinary major risk.

At the biological-survival layer, irreversibility primarily refers to absolute physical or biological nonrecoverability, such as species extinction, rupture of the reproductive chain, or permanent incapacitation.

At the civilizational-conditions layer, irreversibility or difficulty of reversal primarily refers to historical, institutional, and cognitive forms of nonrecoverability, such as institutional path dependence, breakdowns in intergenerational capacity formation, erosion of key judgment capacities, and the failure of a society to recognize the loss of those capacities as a loss.

Here, "difficult-to-reverse" does not mean high governance cost, long recovery time, or difficult institutional reform. It means that, on an intergenerational historical scale, relevant capacities, institutional pathways, and category-calibration mechanisms have deteriorated to the point where realistic agents of repair and realistic pathways of repair are lacking. This paper does not quantify "difficult-to-reverse" by a fixed duration or a single metric. It is a standard of conceptual identification whose concrete operationalization requires further empirical and governance research.

A.29 Humanity as a Future-Oriented Subject

Humanity as a future-oriented subject means that humanity does not merely enter the future passively, but is capable, across generations, of understanding risks, forming judgment, bearing responsibility, organizing action, revising institutions, and shaping the future.

It is not a new risk-bearing subject, nor a concept independent of the realization of subjecthood. It is the temporal and agential dimension of the realization of subjecthood. In other words, humanity as a future-oriented subject means that humanity can direct knowledge, judgment, responsibility, institutional participation, and meaning-creation toward the future, and thereby protect or renew the common world.

This concept is used in the fourth existential-risk threshold: a risk has existential-risk significance only if the harms identified by the first three thresholds converge into overall, intergenerational damage to humanity's capacity as a future-oriented subject.

A.30 Four Thresholds / Existential-Risk Thresholds

Four thresholds are the four standards used in this paper to determine whether a risk to the realization of human subjecthood at the civilizational-conditions layer reaches the level of existential risk.

For a risk to enter the scope of existential risk at the level of the realization of human subjecthood, it must at least:

1. act upon foundational civilizational conditions rather than local functions;

2. have diffusion capacity across domains and civilizational interfaces, rather than being an isolated accident, single misuse, or local reorganization internal to one institutional system;
3. be capable of causing irreversible or difficult-to-reverse structural damage;
4. cause the harms identified in the first three thresholds to converge into overall, intergenerational damage to humanity's capacity as a future-oriented subject.

These thresholds primarily apply to risks to the realization of subjecthood at the civilizational-conditions layer. Species extinction, permanent incapacitation, or the permanent foreclosure of humanity's future directly constitutes existential risk by its own nature and need not be assessed through these four thresholds.

The function of the four thresholds is to prevent conceptual overextension. They ensure that the paper does not elevate all AI-related social impacts into existential risks, and does not misread ordinary technological change as civilizational-level harm.

A.31 Diffusion Capacity across Domains and Civilizational Interfaces

Diffusion capacity across domains and civilizational interfaces refers to the capacity of an AI risk mechanism to spread not only within a single field, industry, state, or institutional system, but through shared interfaces in the modern global civilizational ecology, across multiple functional systems of civilization, and into the knowledge, institutional, technological, and meaning-bearing structures jointly relied upon by multiple civilizational traditions.

Here, "across domains" means that the risk mechanism can cross functional systems of civilization such as education, science, law, media, enterprise, public governance, cultural production, and everyday life. "Across civilizational interfaces" means that the risk mechanism can propagate through global research networks, educational systems, digital platforms, legal-governance interfaces, supply chains, technical standards, and structures of public discourse, producing structural effects across different states, cultures, and civilizational traditions.

This concept is the key scale-filter that distinguishes existential risk to humanity from local social risk. A risk may be extremely serious within a single state or institutional system, yet not necessarily constitute existential risk to humanity. Only when it has diffusion capacity across domains and civilizational interfaces, and further satisfies the other thresholds, can it enter the scope of existential risk at the level of the realization of human subjecthood.

A.32 Ordinary Social Risk

Ordinary social risk refers to risk that affects a local social function, institutional process, industry structure, or technological application. Such risk may be very serious, but it usually has a relatively determinate governance object and path of repair.

For example, the replacement of a profession, the restructuring of an industry, the automation of an institutional process, or errors in an information system may all constitute serious social risks, but they do not necessarily constitute existential risks to humanity.

The difference between ordinary social risk and risk to the realization of human subjecthood lies not in severity alone, but in whether the risk acts upon the foundational civilizational conditions under which humanity can continue to exist as a future-oriented subject, whether it has diffusion capacity across domains and civilizational interfaces, whether it causes irreversible or difficult-to-reverse structural damage, and whether these harms converge into overall, intergenerational damage to humanity's capacity as a future-oriented subject.

A.33 Risk to the Realization of Human Subjecthood

Risk to the realization of human subjecthood refers to risk that acts upon the foundational conditions under which humans realize capacities for knowledge, judgment, responsibility, institutional participation, meaning-creation, and future-directed action under civilizational conditions.

It differs from ordinary social risk. Ordinary social risk usually acts upon local functions, institutional processes, industry structures, or technological applications; risk to the realization of human subjecthood acts upon more foundational civilizational conditions and may affect whether humans can continue to exist as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects.

When such a risk acts upon foundational civilizational conditions, has diffusion capacity across domains and civilizational interfaces, is capable of causing irreversible or difficult-to-reverse structural damage, and causes these harms to converge into overall, intergenerational damage to humanity's capacity as a future-oriented subject, it may enter the scope of AI-related existential risk to humanity.

A.34 Paradigm Extension

Paradigm extension refers to expanding the range of objects or modes of harm recognized by risk identification while preserving the core standing of existing existential-risk frameworks.

The paradigm extension proposed in this paper is not a paradigm replacement. It does not deny species extinction, permanent incapacitation, and future deprivation as baseline forms of existential risk. Rather, it further argues that AI may also constitute existential risk to humanity by undermining, irreversibly or in ways difficult to reverse, the foundational conditions under which humanity realizes subjecthood within civilization.

This concept clarifies the relation between this paper and existing existential-risk theory. The paper does not deny that existing frameworks already include the dimension of future deprivation. Its contribution is to point out that, in the age of AI, future deprivation cannot be understood only as a consequence attached to biological extinction. It may also take the form of humanity continuing to exist while gradually losing the civilizational conditions under which it can understand, judge, take responsibility, and shape the common world as a future-oriented subject.

A.35 Risk-Identification Framework

A **risk-identification framework** is an analytic framework used before specific risk classification, governance design, or technical control. It determines the object of risk, the modes of harm, the thresholds of judgment, and the boundaries of relevant concepts.

This paper provides a risk-identification framework, not a complete taxonomy of risks and not a technical governance proposal. It answers a question prior to all of these: what is AI risk analysis trying to protect? The answer of this paper is that it must protect not only the survival of humanity as a biological species, but also the foundational conditions under which humanity can continue to realize subjecthood within civilization.

The function of a risk-identification framework is to provide comparable, connectable, and criticizable conceptual positions for different discussions of AI risk. It does not replace empirical research, technical governance, or policy design; rather, it provides the prior ontological coordinates for those forms of work.

A.36 AGI Era

The **AGI era**, as used in this paper, does not mean that AGI in the strict sense has already been achieved, nor does it depend on the claim that a universally accepted AGI threshold has been crossed.

It refers to an unfolding historical transition: AI systems are becoming more general, more autonomous, more capable of action, and increasingly deeply embedded in knowledge production, institutional processes, judgment formation, technological research and development, public governance, and meaning generation.

This concept marks a historical condition: even before genuine AGI appears, AI has already begun to change the civilizational conditions under which human subjecthood is realized. Accordingly, many of the risk mechanisms discussed in this paper do not need to wait until a strictly defined AGI threshold is crossed in order to have analytic significance.

A.37 Ontological Coordinate System

An **ontological coordinate system** is the risk-identification framework this paper provides for AI-related existential risk to humanity. By clarifying the risk-bearing subject, the layers of possible harm, the category-calibration mechanism, functional systems of civilization, structures of human participation, AI mechanisms of action, and risk thresholds, it gives different AI risk discussions a structure in which they can be compared, connected, and critically examined.

In this paper, the ontological coordinate system includes:

1. one risk-bearing subject: humanity as an embodied species;
2. two basic layers of possible harm: the biological-survival layer and the civilizational-conditions layer;
3. one internal mechanism: the category-calibration mechanism;
4. one intermediate support layer: functional systems of civilization and structures of human participation;
5. two AI mechanisms of action: deprivation by substitution and mediated reconfiguration;
6. a set of existential-risk thresholds, including diffusion capacity across domains and civilizational interfaces as a scale-filter.

This concept clarifies the relation between this paper and risk taxonomies, technical governance, and policy proposals. The paper does not directly replace those forms of work; it provides the prior conceptual coordinates within which they can be located. What the paper offers is not a complete list of risks, but an ontological coordinate system for judging which kinds of AI-related harm touch the existential foundations of humanity.

V. Structural and Methodological Concept

A.38 Functional Analogy: Hardware–Kernel–Distribution–Application Layer

Functional analogy: hardware–kernel–distribution–application layer refers to the limited analogy used in this paper to help clarify the structural relation among the human species, the philosophical kernel, human civilization, and the functional systems of civilization.

In this analogy, the human species may be analogized to hardware because it provides the most basic biological support; the philosophical kernel may be analogized to the kernel because it functionally maintains the calibration of foundational category interfaces; human civilization may be analogized to a distribution because it constitutes the complete intergenerational operating environment; and the functional systems of civilization may be analogized to the application layer because they are the main interfaces through which humans enter the common world in everyday life.

This analogy explains only functional relations; it does not determine ontological relations. It is not meant to understand humans as machines, philosophy as code, or civilization as software. Its function is to help readers understand the relations among support, interface, operating environment, and application interface, while avoiding two misunderstandings: first, the assumption that the biological survival of the human species is equivalent to the intactness of the conditions for subjecthood-realization; second, the assumption that the continued operation of functional systems of civilization means that the underlying categories remain healthy.

This concept is listed separately in the appendix in order to fix the boundary of the analogy and prevent it from overtaking the argument. What carries argumentative force remains the paper's non-analogical structural account of the relation among the human species, human civilization, the philosophical kernel, and the functional systems of civilization.

Notes

[Note 1]

The "AGI era" in this paper does not presuppose that AGI in the strict sense has already been achieved, nor does it depend on the claim that a generally accepted AGI threshold has been crossed. It refers to an unfolding historical transition: AI systems are becoming more general, more autonomous, more capable of action, and increasingly deeply embedded in knowledge production, institutional processes, judgment formation, technological research and development, public governance, and meaning generation. Several risk mechanisms discussed in this paper may already appear before genuine AGI arrives, in the form of advanced AI, generative AI, agentic AI, and automated systems.

[Note 2]

This paper treats AI as a civilizational mediator that may reorganize the structures of human knowledge, action, responsibility, and meaning. This problem-setting is adjacent to traditions of technological mediation in the philosophy of technology and postphenomenology. For relevant background, see Heidegger (1977), Ihde (1990), Winner (1980), Verbeek

(2011), Rosenberger and Verbeek (2015), and Pezzano and Pavanini (2025). The deeper background concerning the common world, action, and responsibility for the future may also be read in relation to Arendt (1958) and Jonas (1984).

[Note 3]

Existing discussions of existential risk and AI risk already cover species extinction, permanent incapacitation, future deprivation, loss of control, the control problem, and irreversible loss of long-term future potential. See Bostrom (2002, 2013, 2014), Ord (2020), Russell (2019), and Carlsmith (2022). Recent philosophical and legal scholarship has also begun systematically to classify AI existential risk; see Cappelen, Goldstein, and Hawthorne (2025), and Tokson and Arbel (2026). This paper is adjacent to these studies, but it does not propose a new risk taxonomy or regulatory scheme. It instead clarifies, at a more prior level, the "humanity" that bears these risks and the ways in which it may be harmed.

[Note 4]

The paper's conceptual-engineering orientation is related to discussions of conceptual revision, conceptual design, and normative conceptual analysis in Cappelen (2018), Chalmers (2025), and Haslanger (2000). This paper draws on the problem-setting of conceptual engineering, but its object is not an internal debate in the philosophy of language. Its object is the problem of the risk-bearing subject in AI risk theory: when we say that AI threatens "humanity," what exactly is the "humanity" at stake?

[Note 5]

The distinction between formal participation and substantive participation is not meant to deny the importance of procedural participation. It is meant to point out that retaining a procedural position, final signature, or formal human-in-the-loop structure does not necessarily mean that humans still understand the formation of judgment, bear substantive responsibility, or shape institutional direction. Substantive participation, as used in this paper, emphasizes that humans, as knowers, judges, bearers of responsibility, maintainers of institutions, creators of meaning, and future-oriented subjects, exert real influence on the operation and renewal of the common world. On the problem of responsibility being wrongly compressed onto human actors with limited control in complex automated systems, see Elish (2019) on moral crumple zones. On the responsibility gap in learning automata, see the classic analysis by Matthias (2004).

[Note 6]

The "intergenerational common world" in this paper is not directly identical with any single concept in political philosophy or phenomenology. It is, however, problematically related to Arendt's discussions of the common world, action, and the human condition, and to

Jonas's discussions of responsibility in the technological age and responsibility for future generations. What this paper draws from these discussions is a problem-setting: human civilization is not the simple continuation of population through time, but the historical condition that enables the past to be transmitted to the future and enables humans to understand, judge, take responsibility, and shape the common world.

[Note 7]

This paper's discussion of truth, evidence, knowledge communities, public judgment, and information ecologies is connected to the tradition of social epistemology. For relevant background, see Goldman (1999), Longino (1990), Fricker (2007), O'Connor and Weatherall (2019), and Nguyen (2020). More directly, Ortmann (2025) discusses epistemic dependence on AI in science and argues that the opacity of AI tools does not necessarily pose novel problems for social epistemology. This paper is adjacent to that discussion, but operates at a different level: it is not primarily concerned with whether AI opacity constitutes a new problem for social epistemology, but with whether mediated reconfiguration may touch the foundational conditions for the realization of human subjecthood when AI deeply enters the functional systems of civilization and changes category interfaces such as truth, evidence, responsibility, subjecthood, value, and meaning.

[Note 8]

The hardware–kernel–distribution–application-layer analogy is used only as a functional explanation, not as an ontological model. It does not imply that human civilization can be modularly decomposed, replaced, reinstalled, or migrated like a computing system, nor that the philosophical kernel is an entity-module that can be separately installed, replaced, or restarted. The analogy is used only to explain four functional relations: the human species provides biological support; the philosophical kernel calibrates foundational category interfaces; human civilization provides a complete intergenerational operating environment; and the functional systems of civilization provide the application-layer interfaces through which humans actually enter the common world.

[Note 9]

This paper does not claim that AI necessarily weakens the conditions for the realization of human subjecthood. AI may also enhance education, scientific research, public services, medical diagnosis, accessibility support, access to knowledge, institutional transparency, and public collaboration. The paper concerns another possibility: when the way AI is embedded acts upon foundational civilizational conditions, has diffusion capacity across domains and civilizational interfaces, may cause irreversible or difficult-to-reverse structural damage, and converges into overall, intergenerational damage to humanity's capacity as a future-oriented subject, then it enters the scope of existential risk at the level of the realization of human subjecthood as defined in this paper.

[Note 10]

On the possibility that AI risk may occur through the gradual weakening of humanity's influence over large-scale social systems such as the economy, culture, and the state, see Kulveit et al. (2025), "Gradual Disempowerment: Systemic Existential Risks from Incremental AI Development." In the coordinate system of this paper, such work can be located within the problem of functional systems of civilization and structures of human participation at the civilizational-conditions layer, and is especially close to what this paper calls deprivation by substitution. This paper adopts its risk intuition concerning gradual disempowerment and power transfer, but does not take "power transfer" itself as its final conceptual center.

[Note 11]

On the relation between the use of generative AI and critical thinking, cognitive effort, and modes of task supervision, see the survey of knowledge workers by Lee et al. (2025). On students' over-reliance on AI dialogue systems and its effects on cognitive abilities, see the systematic review by Zhai, Wibowo, and Li (2024). These studies do not directly prove existential risk, but they provide empirical background for the mediated reconfiguration discussed in this paper: when generative AI or AI dialogue systems are embedded in knowledge work and learning processes, human practices of critical thinking, cognitive effort, burdens of judgment, and modes of task supervision may be redistributed.

[Note 12]

For the classic statement of the extended mind, see Clark and Chalmers (1998). On whether large language models can be understood as part of the extended mind, see Smart, Clowes, and Clark (2025) on LLMs and the extended mind. This paper is adjacent to that discussion, but it does not center on the question whether LLMs constitute an extended mind. It asks instead whether, when AI systems organize points of information entry, reason generation, option ranking, and the environment of judgment, humans still retain the substantive conditions for forming judgment, bearing responsibility, and realizing subjecthood.

[Note 13]

"Diffusion capacity across domains and civilizational interfaces" in this paper does not mean that a risk must affect all states or civilizational traditions at the same time and in the same way. It means that its pathways of propagation have entered the shared interfaces of the modern global civilizational ecology, such as global research networks, educational systems, digital platforms, legal-governance interfaces, supply chains, technical standards, and structures of public discourse. The function of this threshold is to prevent serious social risks within a single state, industry, or institutional system from being directly elevated into existential risks to humanity.

[Note 14]

At the civilizational-conditions layer, "irreversible or difficult to reverse" primarily refers to historical, institutional, and cognitive difficulty of recovery, rather than a single form of physical irreversibility. This discussion can be placed in dialogue with research on path dependence, institutional persistence, and capacity formation—for example, Arthur (1989) on technological lock-in and increasing returns, Pierson (2000) on path dependence in politics, North (1990) on institutional change, and Cunha and Heckman (2007) on the technology of skill formation. This paper does not quantify "difficult to reverse" by a fixed duration or a single metric. It does not mean high governance cost or long recovery time. It means that, on an intergenerational historical scale, relevant capacities, institutional pathways, and category-calibration mechanisms have deteriorated to the point where realistic agents of repair and realistic pathways of repair are lacking.

[Note 15]

On the possibility that AI may bring deeper risks by weakening human autonomy, decision-making capacity, and skill practice, see Prunkl (2022) on human autonomy in the age of AI, and Krook (2025) on autonomy decline, skill degradation, and existential risk. In the coordinate system of this paper, these discussions can be located within the problem of weakened capacities of subjecthood at the civilizational-conditions layer, and they intersect with both deprivation by substitution and mediated reconfiguration. The main axis of this paper is not autonomy theory itself, nor any single form of skill degradation, but the question of when such declines in capacity rise to the level of existential harm to humanity as the risk-bearing subject.

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